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INGENIERÍA INFORMÁTICA**

Bachelor's Thesis

Analysis and Modelling of the Impact of Poland's Air and Economic Growth on European Tourism Flows

DEGREE: Bachelor's Degree in Data Science and Engineering

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Summary

This Bachelor's Thesis models and forecasts international tourism demand in Poland through a methodology of progressive complexity. By integrating macroeconomic, demographic and air connectivity variables drawn mainly from Eurostat and the International Monetary Fund, a comparative battery of predictive models is constructed spanning classical time series methods (SARIMA, SARIMAX), machine learning (Random Forest, XGBoost) and recurrent deep learning (LSTM with exogenous variables). After systematically addressing the endogeneity of air capacity, the models are evaluated through temporal validation with expanding-window backtesting and formal significance tests. The work is completed with two applied components: a forward-looking pipeline fed by International Monetary Fund projections that simulates demand under alternative economic scenarios, and a causal analysis of the COVID-19 impact using Bayesian structural time series. The result is an integrated strategic planning tool with operational value for airlines and destination managers.

Keywords: tourism demand, time series forecasting, Poland, LSTM, SARIMAX, XGBoost, Bayesian statistics, Eurostat, air connectivity, COVID-19, IMF scenarios.

Abstract

This Bachelor's Thesis models and forecasts international tourism demand in Poland through a methodology of progressive complexity. By integrating macroeconomic, demographic and air connectivity variables mainly from Eurostat and from the International Monetary Fund, the work develops a comparative battery of predictive models spanning classical time series (SARIMA, SARIMAX), machine learning (Random Forest, XGBoost) and recurrent deep learning (LSTM with exogenous variables). After systematically addressing the endogeneity of air capacity, the models are evaluated through temporal validation with expanding-window backtesting and formal tests of significance. The work is completed with two applied components: a forward-looking pipeline fed by International Monetary Fund projections that simulates demand under alternative economic scenarios, and a causal analysis of the COVID-19 impact using Bayesian structural time series. The result is an integrated strategic planning tool with operational value for airlines and destination managers.

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Chapter 1

Introduction

1.1 International tourism as a socioeconomic phenomenon

International tourism is one of the most significant economic sectors at the global scale. According to the World Tourism Barometer of the United Nations World Tourism Organization, international arrivals reached 1.4 billion in 2024, recovering almost entirely to pre-COVID-19 pandemic levels. The World Travel and Tourism Council estimated that the sector's total contribution to global Gross Domestic Product was USD 10.9 trillion in 2024—around 10 % of the global economy—sustaining 357 million direct and indirect jobs. Within Europe, tourism contributed approximately EUR 1.8 trillion to the GDP of the European Union, consolidating itself as one of the continent's structural engines (WTTC, 2025).

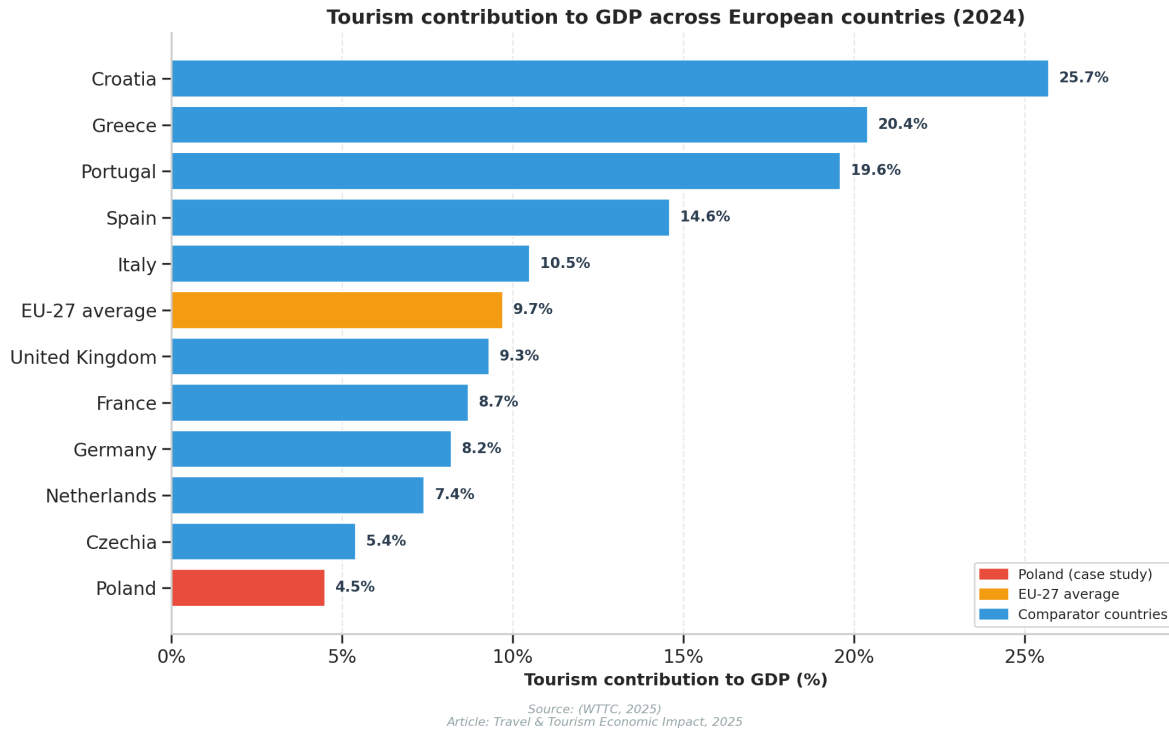


Figure 1.1: Tourism's contribution to GDP in selected European countries, 2024.

The heterogeneity shown in Figure 1.1 reflects the structural diversity of European tourism: tourism-intensive economies—where the sector exceeds 20 % of GDP—coexist with emerging economies where tourism still represents a moderate fraction of aggregate activity. Poland belongs to this second group, with a still considerable margin for growth that makes it particularly interesting from the perspective of forecasting and strategic planning.

1.2 The importance of forecasting tourism demand

The ability to anticipate tourism demand with precision is of first-order strategic importance. Tourism products are intensely perishable: an unoccupied hotel room on a given night or an unsold airline seat on a specific flight represent revenue that is irreversibly lost. Anticipatory forecasting improves the allocation of scarce resources, shaping the planning of airport infrastructure, the scheduling of air routes, the management of accommodation capacity, the sizing of seasonal workforces, and the coordination of the entire tourism value chain.

The economics literature classifies international tourism as a luxury good: the meta-analysis by Peng, Song, Crouch and Witt (2015), based on 194 elasticity studies, places the average income elasticity at around 2.0, confirming that tourism demand is highly sensitive to macroeconomic fluctuations and thus substantially exposed to economic cycles and structural shocks.

This volatility was dramatically exposed during the COVID-19 pandemic, the largest known collapse in the modern history of international tourism. According to Eurostat, overnight

stays by foreign tourists in the European Union fell by around 85 % in April 2020 compared to the same month the previous year, and the series took approximately 28 months to recover to pre-pandemic levels (Eurostat, 2025a). This episode reinforced the need for robust predictive tools capable of integrating not only the sector’s endogenous temporal dynamics but also the macroeconomic, demographic and connectivity determinants that shape tourism flows.

1.3 Poland as a case study

This work takes international tourism in Poland as its case study, a choice driven by several converging factors. Since its accession to the European Union on 1 May 2004, Poland has undergone an unprecedented economic and structural transformation in recent European history. Its real GDP per capita, measured in chained volumes at 2010 reference prices, more than doubled over the period 2004–2024, rising from approximately EUR 7,300 to EUR 15,750 per inhabitant (a growth factor of 2.16) (Eurostat, 2025b; Eurostat, 2025c). Its price level, as measured by the Harmonised Index of Consumer Prices, remains notably below the EU average, maintaining a considerable competitive advantage in terms of purchasing power parity for foreign visitors.

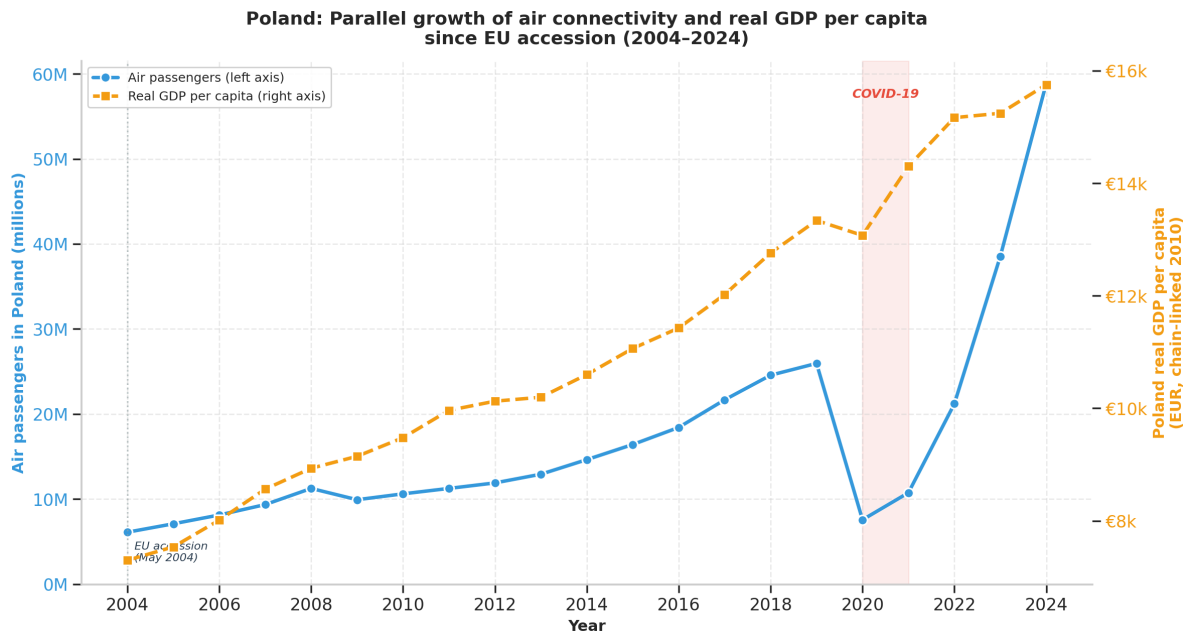


Figure 1.2: Parallel growth of air traffic and real GDP per capita in Poland, 2004–2024.

Figure 1.2 captures the central macroeconomic narrative of this work: Poland’s economic transformation has been accompanied by a spectacular expansion of air connectivity. Over the same period in which its real GDP per capita doubled, the number of passengers at Polish airports rose from 6.1 million in 2004 to 58.9 million in 2024, a growth of nearly ten-fold. Both curves grow in parallel and are simultaneously affected by the pandemic shock in 2020, suggesting a structural link between macroeconomic performance and air connectivity

dynamics that this work sets out to formally model. This expansion has been driven to a large extent by the massive penetration of low-cost carriers, particularly Ryanair and Wizz Air, which have transformed the structure of the Polish air market by opening direct routes to virtually all major Western European source markets.

In 2024, Poland recorded the second highest tourism growth rate in the entire European Union—surpassed only by Malta—with a year-on-year increase of close to 6 % in international overnight stays. The combination of economic dynamism, price competitiveness, expanding air connectivity and sustained tourism growth makes Poland an ideal laboratory for the quantitative analysis of the interaction between macroeconomic variables, air connectivity and tourism demand.

1.4 General objective and dual orientation of the work

The general objective of this work is to understand, quantify and forecast the interaction between Poland’s macroeconomic, demographic and air connectivity indicators and its international tourism demand, through the development of predictive models of progressive complexity grounded in time series analysis, machine learning, deep learning and Bayesian statistics.

The work presents a **dual orientation** that distinguishes it from most works in the area. First, it has an **academic** component: it systematically explores the applicability of statistical and artificial intelligence methods to the specific problem of Poland’s aggregate tourism demand, comparing their performance rigorously and reproducibly, and documenting the endogeneity and collinearity problems that affect this class of models. Second, it incorporates a **market-applied** component: the work culminates in prospective tools that allow tourism demand and air capacity forecasts to be conditioned on future macroeconomic projections from international organisations—specifically, the *World Economic Outlook* of the International Monetary Fund—under alternative scenarios.

This applied orientation is not incidental. The work includes a specific module, developed in dialogue with an aviation sector economist, aimed at forecasting the seat capacity offered at Polish airports from macroeconomic inputs, with a deliberately interpretable formulation designed for use in the strategic planning of an airline. The distinction between what is optimal for academic inference and what is optimal for operational decision-making—where the constraints on data availability and model interpretability are as important as precision—is one of the original conceptual contributions of the present work and will be developed in detail in the following chapters.

1.5 Structure of the thesis

The rest of the document is organised according to the structure of the School of Computer Engineering. **Chapter 2** describes the current state of tourism demand modelling, the methodological background and the specific objectives of the work. **Chapter 3** details the contributions of the work, the specific competences of the Bachelor’s Degree in Data Science

and Engineering that are addressed, and the alignment with the Sustainable Development Goals. **Chapter 4** sets out the development: the CRISP-DM methodology adopted, the integration and treatment of data sources, feature engineering, the Endogeneity diagnostic and the presentation of the model battery. **Chapter 5** presents and discusses the results of all models, including the rigorous comparative assessment, the prospective module under IMF scenarios, the causal analysis of COVID-19 and the applied model for air capacity forecasting. **Chapter 6** contains the conclusions, the degree to which the objectives have been achieved, lines for future work and the declaration of artificial intelligence use. The document closes with the bibliography and the glossary of terms.

Chapter 2

Current state and initial objectives

2.1 Motivation and background

2.1.1 Conceptual framework of tourism demand

The econometric modelling of tourism demand has a consolidated tradition dating back to the 1970s. The works of Crouch (1994, 1995) established the set of fundamental determinants that, with variations, has guided research during the three following decades. These classical determinants can be grouped into four blocks.

The first block consists of **income variables from source markets**. Microeconomic theory predicts that international tourism, being a good with an income elasticity greater than one, should exhibit a positive and elastic relationship with the GDP or disposable income of origin countries. The meta-analyses of Crouch (1995) and subsequently of Peng, Song, Crouch and Witt (2015) confirm this prediction empirically, with average income elasticities ranging between 1.5 and 2.5.

The second block corresponds to **relative prices**. When choosing between alternative destinations, tourists compare both the relative cost of living between the source and destination country (differential inflation) and bilateral exchange rates, which determine the effective purchasing power of their currency at the destination. The most influential operationalisation of tourist prices was provided by Song, Li, Witt and Fei (2010) through the formulation of the **relative tourist price**:

$$P_{it} = \frac{CPI_{PL,t}}{CPI_{i,t}} \times \frac{EX_{i,t}}{EX_{PL,t}} \quad (2.1)$$

where CPI_{PL} and CPI_i represent the Harmonised Indices of Consumer Prices for the destination (Poland) and source country i , and EX_{PL} and EX_i denote the bilateral exchange rates against the euro. This variable simultaneously captures the cost-of-living differential and the effect of currency fluctuations. Its effective construction for the Polish case is detailed in Chapter 4.

The third block consists of **economic sentiment and consumer confidence indicators**, of a leading nature (*leading indicators*): they capture expectations about future economic developments that anticipate discretionary spending decisions, including tourism. The Consumer Confidence Indicator published by the European Commission constitutes the standard harmonised reference.

The fourth block, less common in the classical literature but central to the present work, is that of **air connectivity variables**. The effective availability of seats on direct flights from source markets constitutes a supply constraint that conditions observed demand, and has gained increasing relevance in the last decade with the expansion of the low-cost model in Europe.

2.1.2 The evolution of low-cost carriers in Poland

Poland's tourism growth cannot be separated from the structural transformation of its air market, progressively dominated by low-cost carriers (LCC). These grew from representing less than 5 % of the Polish market in 2004 to more than 52 % by 2015 (Huderek-Glapska and Nowak, 2016), and have continued to gain market share, exceeding 59 % in 2023. This phenomenon results from the combination of a considerable latent demand from Polish diasporas in Western Europe—particularly in the United Kingdom, Germany and the Netherlands following the large migration waves after 2004—and an aggressive expansion strategy by Ryanair and Wizz Air, which established operating bases in Warsaw, Kraków, Gdańsk and Katowice. Figure 2.1 shows the composition of air traffic to Poland broken down by the nine main source markets, where Spain, Germany and the United Kingdom clearly appear as the three largest with shares of around 19–20 % each.

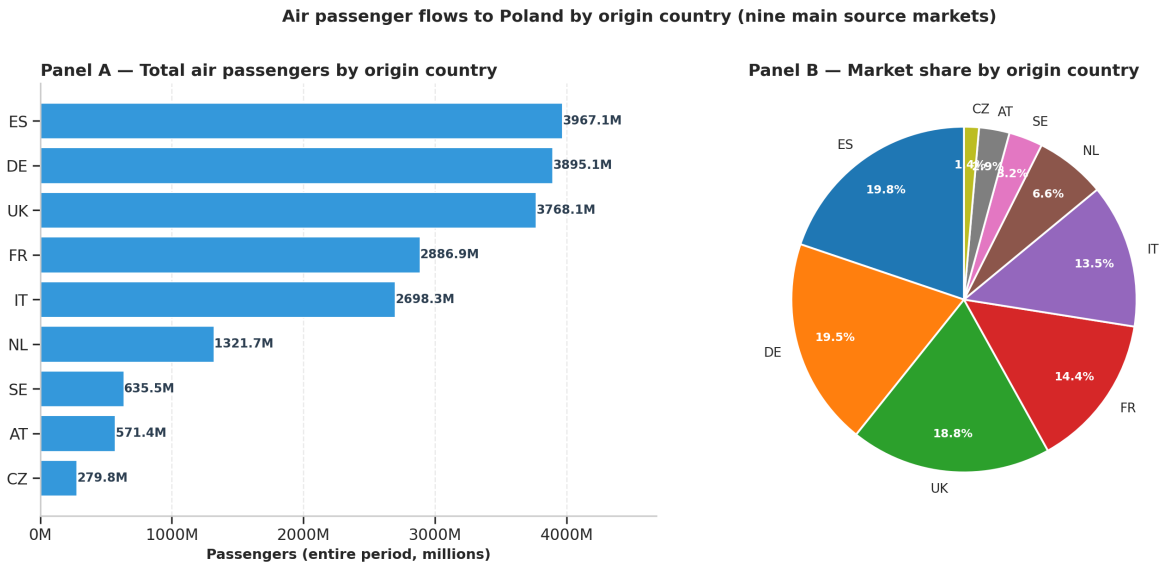


Figure 2.1: Air traffic to Poland by country of origin: total volumes (Panel A) and market shares (Panel B).

2.1.3 Foreign tourist overnight stays as the target variable

The target variable of the work is the monthly overnight stays by foreign tourists in tourist accommodation establishments in Poland, recorded by Eurostat in dataset `tour_occ_nim` (Eurostat, 2025a) in accordance with Regulation (EU) 692/2011 on European tourism statistics. The choice is justified by three considerations. First, overnight stays are a direct measure of tourism’s economic impact on the destination, superior to alternative metrics such as arrivals (which do not capture length of stay) or revenues (subject to recording and cross-country harmonisation problems). Second, monthly frequency allows the capture of fine seasonal dynamics and short-term sensitivity, unlike the annual series that are commonly used, which cannot be sufficiently precise. Third, the European harmonisation under Regulation 692/2011 ensures data comparability and robustness. Figure 2.2 shows these monthly overnight stays by foreign tourists in Poland, highlighting the most relevant events of recent years.

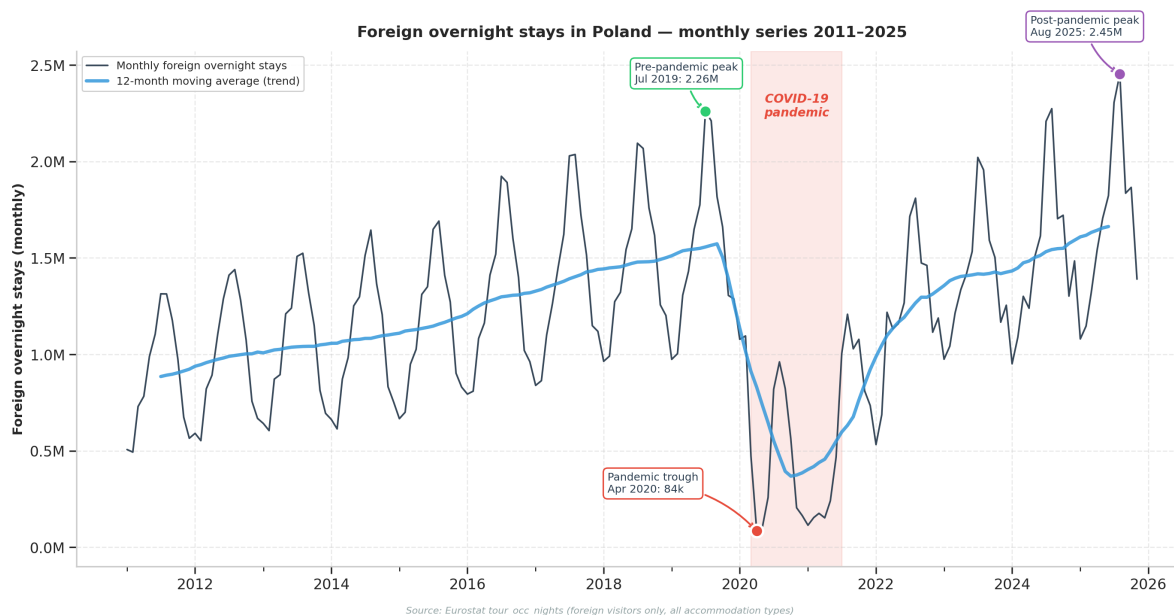


Figure 2.2: Monthly overnight stays by foreign tourists in Poland, 2011–2025, with pre-pandemic milestones, COVID-19 trough and post-pandemic peak.

A relevant limitation of the `tour_occ_nim` dataset (Eurostat, 2025a) conditions the design of the work and must be documented explicitly: the country of residence variable only distinguishes three aggregate categories—domestic residents, foreigners and total—and does not provide a breakdown by country of origin. The work therefore models the aggregate series of foreign overnight stays, incorporating the dimension of source markets through exogenous variables measured in each country of origin, weighted by the air passenger shares derived from dataset `avia_paoc` (Eurostat, 2025d).

2.1.4 Methodological landscape: from classical econometrics to deep learning

The forecasting of tourism demand has undergone an accelerated methodological evolution over the past two decades. Song, Qiu and Park (2019), in a review of 214 studies published between 1968 and 2018, identify three successive waves: classical univariate econometrics (ARIMA and variants), multivariate econometrics with exogenous variables (SARIMAX, VAR, VECM) and the current wave of machine learning and deep learning.

In the first wave, models from the ARIMA family and Holt-Winters exponential smoothing dominated practice until approximately 2010. Athanasopoulos, Hyndman, Song and Wu (2011), in a systematic tourism forecasting competition, documented that these classical models, despite their simplicity, are difficult to beat consistently, especially at short horizons and with strongly seasonal series.

The second wave incorporated exogenous variables into the autoregressive structure. A recurring and relevant finding for this work is that documented by Gunter and Smeral (2016): elasticities estimated via multivariate models tend to be smaller than those estimated with simpler univariate models. The authors attribute this to the progressive absorption of tourism variation by autocorrelated components, and warn against over-interpreting elasticities derived from linear models saturated with regressors. This warning, explicitly noted by the supervisor of the present work, will be empirically confirmed in Chapter 5.

The third wave corresponds to machine learning and deep learning. Studies such as Salamanis, Xanthopoulou, Kehagias and Tzovaras (2022) or Chen, Ying, Zhang and Balezentis (2024) document that recurrent neural networks, particularly LSTM (*Long Short-Term Memory*) architectures, can significantly reduce forecasting errors at long horizons when sufficient training data are available. However, Hewamalage, Bergmeir and Bandara (2021), in a comprehensive evaluation of recurrent networks for time series, warn that these architectures are sensitive to structural breaks and require considerable sample sizes; for short series with regime changes, classical and tree-based models tend to be competitive or superior. This nuance is central to the present work, where the target series is relatively short (179 monthly observations) and contains the COVID-19 structural shock.

In parallel, Bayesian methods have emerged as a robust probabilistic alternative for causal inference. The *CausalImpact* framework developed by Brodersen, Gallusser, Koehler, Remy and Scott (2015), based on Bayesian structural time series models, allows the causal counterfactual impact of an exogenous intervention—such as the COVID-19 pandemic—to be estimated by constructing a synthetic series that estimates what the evolution would have been in the absence of that intervention. This approach is applied in the present work to rigorously quantify the effect of COVID-19 on Polish tourism.

2.1.5 The problem of Endogeneity of air capacity

A methodological problem running throughout the work, identified during exploratory analysis and addressed systematically, is the **endogeneity of air capacity**. Economic intuition suggests that the number of seats offered conditions observed tourism demand; but the reverse

is also true: airlines schedule capacity in anticipation of expected demand. There is therefore a relationship of reciprocal causation between capacity and demand. The naive inclusion of contemporaneous air capacity as a predictor of demand produces models with an apparently excellent but misleading fit, in which the importance of macroeconomic variables is artificially masked. The formal diagnosis of this endogeneity—through bidirectional Granger causality tests—and its treatment through a scheme of specifications of decreasing complexity constitute one of the methodological contributions of the work, developed in Chapters 4 and 5.

2.2 Objectives

2.2.1 General objective

The general objective of this work is to analyse and forecast the evolution of international tourism associated with Poland by integrating economic, demographic and air capacity data, constructing models based on time series, machine learning, deep learning and Bayesian statistics, and evaluating their comparative performance under different specifications and forecasting horizons.

2.2.2 Specific objectives

The specific objectives set out in the initial proposal are articulated in the following points.

First. To carry out a descriptive and comparative analysis of the evolution of air seats, real and nominal GDP, population and the main tourism metrics in Poland relative to its Western European source markets.

Second. To calculate key derived indicators, such as travel propensity, air seats per capita, ratios between economic growth and air connectivity and, in particular, the relative tourist price of Song et al. (2010) weighted by the air traffic shares of the source markets.

Third. To model and quantify the impact of each variable—macroeconomic, demographic and air connectivity—on tourism flows to Poland, recognising the feedback loop between economic growth, air traffic and tourism expenditure documented in the area’s literature.

Fourth. To develop a progressive battery of predictive models including:

- ✓ Classical models: ARIMA, SARIMA and regression with exogenous variables (SARIMAX);
- ✓ Machine learning algorithms: Random Forest and XGBoost;
- ✓ Deep learning models for time series: recurrent LSTM networks with exogenous variables, in different architectural configurations.

Fifth. To generate forecasts under alternative scenarios—such as a baseline, an optimistic and a pessimistic scenario regarding the macroeconomic evolution of source markets and Poland’s air capacity—and to evaluate their impact on European tourism. This objective

explicitly includes the development of an *Airline Capacity Prediction* module that forecasts offered seat capacity from macroeconomic inputs, with direct utility for strategic planning in the aviation sector.

2.2.3 Deviations from the initial proposal

The development of the work has adhered substantially to the initial proposal, keeping all five specific objectives intact. The most relevant deviation from what was planned is methodological rather than one of scope: the treatment of the structural change caused by COVID-19 revealed, at an intermediate stage of the work, the advisability of adopting a formulation of the recurrent model's target as year-on-year growth rates rather than absolute levels. This decision, not anticipated in the initial proposal, turned out to be one of the most relevant methodological contributions of the work and is developed in detail in Chapter 5. Additionally, whether ultimately determinant by their results or not, the evaluation methodology incorporated formal significance tests—specifically the Diebold-Mariano test with the Harvey-Leybourne-Newbold correction for small samples—and an expanding-window backtest scheme at multiple horizons, methodological refinements not specified in the initial proposal but aligned with the intention of rigorously evaluating the model battery.

Chapter 3

Contributions of the work

This chapter justifies the value that the work contributes to its socioeconomic, technical and scientific environment, indicates which specific competences of the Bachelor’s Degree in Data Science and Engineering have been developed, and discusses the project’s alignment with the Sustainable Development Goals.

3.1 Main contributions

The work adds value both to the technical domain of data science and to the applied analysis of the tourism sector. Its main contributions are grouped under six axes.

Multidimensional integration and harmonisation of official sources. A master monthly panel is consolidated that resolves historical shortcomings in the integration of heterogeneous sources: Eurostat for tourism, macroeconomic and air connectivity variables, the Office for National Statistics (ONS) of the United Kingdom for series that ceased to be published in a harmonised form after Brexit, and International Monetary Fund projections for the prospective module. The treatment of the United Kingdom by means of statistical *Chain-linking (statistical chain-linking)* over the overlap period with Eurostat is a substantial improvement over the usual approaches of extrapolation or exclusion of the British market, which accounts for around 19 % of air traffic to Poland.

Rigorous multi-paradigm comparison. The work empirically quantifies the marginal value of four methodological paradigms—classical econometrics, machine learning, deep learning and Bayesian statistics—on the same problem and with a common evaluation protocol. Unlike superficial comparisons based on a single Hold-out, the evaluation combines temporal cross-validation with Expanding-window backtest at multiple horizons and a Diebold-Mariano test with correction for small samples, providing robust evidence on the circumstances in which each paradigm adds incremental value.

Systematic diagnostic of Endogeneity and collinearity. The work develops a diagnostic framework that identifies, documents and systematically addresses the problems of endogeneity—fundamentally the reciprocal causation between air capacity and tourism

demand—and of collinearity—between weighted aggregates and individual components, between trending variables sharing a common stochastic trend—that typically affect tourism demand models. The progression of specifications of decreasing complexity, from a naive to a purely macroeconomic one, allows the marginal contribution of each informational block to be isolated and constitutes an original methodological contribution.

Transformation to year-on-year growth rates as a response to structural change.

The work empirically identifies that the seasonal shape of Polish overnight stays is stable across the COVID-19 shock (correlations between pre- and post-pandemic seasonal patterns above 0.94) while the absolute level changed structurally ($\sim 40\%$ higher post-2024). It exploits this decomposition to reformulate the recurrent model’s target as the year-on-year ratio $\text{nights}[t]/\text{nights}[t-12]$ rather than absolute levels, thereby neutralising the structural change and allowing the network to learn only the stable seasonal dynamics. This design decision produces improvements of 42 % to 83 % in forecast accuracy relative to the levels formulation, and constitutes a methodological contribution with potential for generalisation to other series with structural breaks.

Conceptual distinction between academic and applied modelling. The work explicitly articulates a distinction that, although implicit in much of the literature, is rarely clearly formulated: the optimal specification of a model depends on the objective. For academic inference on the determinants, short-term autoregressive lags are informative and should be retained; for operational prediction with information available at the time of the decision, those same lags are useless because they represent information not available when the forecast is needed. This distinction justifies the coexistence of two specifications of the same model—an “academic” and an “operational” one—and constitutes a relevant conceptual contribution to the practice of tourism *forecasting*.

Applied tools oriented towards the market. Unlike the univariate forecasts typical of the literature, the work provides two conditional systems with direct operational utility. The first translates external macroeconomic forecasts, such as the International Monetary Fund’s *World Economic Outlook*, into tourism demand forecasts under alternative scenarios. The second, developed in dialogue with an economist with experience in the aviation sector, forecasts offered seat capacity from macroeconomic inputs, with a deliberately interpretable formulation oriented towards strategic planning. Both modules reinforce the applied dimension of the work and constitute simulation tools with direct utility for airlines, tour operators and destination managers.

3.2 Specific competences

The work develops and consolidates the specific competences of the Bachelor’s Degree in Data Science and Engineering listed in the initial proposal. Below, compliance with each of them is justified with direct reference to the phases of the work in which they are materialised.

ED1. Ability to know the foundations, paradigms and techniques of intelligent systems and to analyse, design and build computer systems, services and applications that use such techniques in any application domain. This competence

is evidenced by the knowledge and practical application of four distinct intelligent systems paradigms—classical econometrics, tree-based machine learning, recurrent deep learning and Bayesian statistics—on the same applied problem in the domain of tourism and air connectivity. The design of the complete *pipeline*, from data integration to the prospective module of Chapter 5, exemplifies the comprehensive scope of this competence.

ED3. Ability to know and develop computational learning techniques and to design and implement applications and systems that use them, including those dedicated to the automatic extraction of information and knowledge from large volumes of data. This competence is covered by the practical implementation of six families of computational learning models, including hyperparameter tuning, temporal validation with expanding-window backtesting and an empirical comparison with formal significance tests. Knowledge extraction is evidenced both in the identification of relevant macroeconomic relationships and in the empirical discovery of the value of the year-on-year formulation in recurrent networks.

ED4. Ability to identify and analyse problems and to design, develop, implement, verify and document software solutions in application domains of Artificial Intelligence in Data Science and Engineering. This competence is evidenced by the complete cycle of the work: early identification of the air capacity Endogeneity problem, design of the scheme of four specifications of decreasing complexity as an empirical response to the problem, reproducible implementation in Jupyter notebooks, verification through formal statistical tests and detailed documentation both in code and in the present thesis.

ED6. Ability to have an in-depth knowledge of the fundamental principles and models used in Data Science, particularly those related to the analysis, prediction and forecasting of large volumes of data. This is the competence most directly aligned with the nature of the work, which combines multivariate exploratory analysis, forecasting through a battery of models and prospective analysis conditioned by external IMF scenarios, all on an integrated panel of macroeconomic and tourism data.

ED9 (Principal). Ability to define within the organisation problems in the domain of Data Science and Engineering and to translate statistical analyses into data-driven Business Intelligence actions aimed at improving performance. This competence is materialised in an especially clear form in the work’s applied modules through prospective pipelines, which translate IMF macroeconomic projections into actionable tourism demand forecasts. Additionally, a model designed in dialogue with an economist with experience in the aviation sector and delivered as a closed-form interpretable formula, directly usable for airline capacity planning, is also used. Both modules exemplify the translation of statistical analyses into effective Business Intelligence.

3.3 Alignment with the Sustainable Development Goals

Table 3.1 and Figure 3.1 summarise the degree of relationship of the work with each of the seventeen United Nations Sustainable Development Goals, in accordance with the School’s template criteria.

Table 3.1: Degree of relationship of the work with the Sustainable Development Goals.

SDGs	Degree	Justification
1. No poverty	0	Not directly applicable.
2. Zero hunger	0	Not applicable.
3. Good health and well-being	0	Not applicable.
4. Quality education	1	The project is published openly and constitutes reproducible material that can be used for teaching purposes.
5. Gender equality	0	Not applicable.
6. Clean water and sanitation	0	Not applicable.
7. Affordable and clean energy	0	Not applicable.
8. Decent work and economic growth	3 (High)	Tourism sustains a substantial fraction of employment in Poland and in the source markets considered. Accurate predictive models mitigate seasonal labour instability, enable earlier planning of temporary hirings and optimise hotel and transport capacity management.
9. Industry, innovation and infrastructure	3 (High)	The explicit integration of air capacity into the modelling framework underpins long-term investment decisions on airport infrastructure, route planning and slot allocation.
10. Reduced inequalities	0	Not applicable.
11. Sustainable cities and communities	2 (Medium)	Anticipating demand peaks enables destination cities to plan urban public transport, avoid seasonal congestion (<i>overtourism</i>) and appropriately size seasonal services.
12. Responsible consumption and production	2 (Medium)	Optimising the use of perishable resources—empty aircraft seats, unoccupied hotel rooms, unsold slots—improves the overall efficiency of the system and reduces the waste associated with over-dimensioning.
13. Climate action	0	Not directly applicable, although better capacity planning may indirectly reduce emissions from over-dimensioning.
14. Life below water	0	Not applicable.
15. Life on land	0	Not applicable.
16. Peace, justice and strong institutions	0	Not applicable.
17. Partnerships for the goals	1 (Low)	Exclusive use of open data and auditable public sources (Eurostat, ONS, IMF), publication of the code in a public repository and commitment to open science and reproducibility.

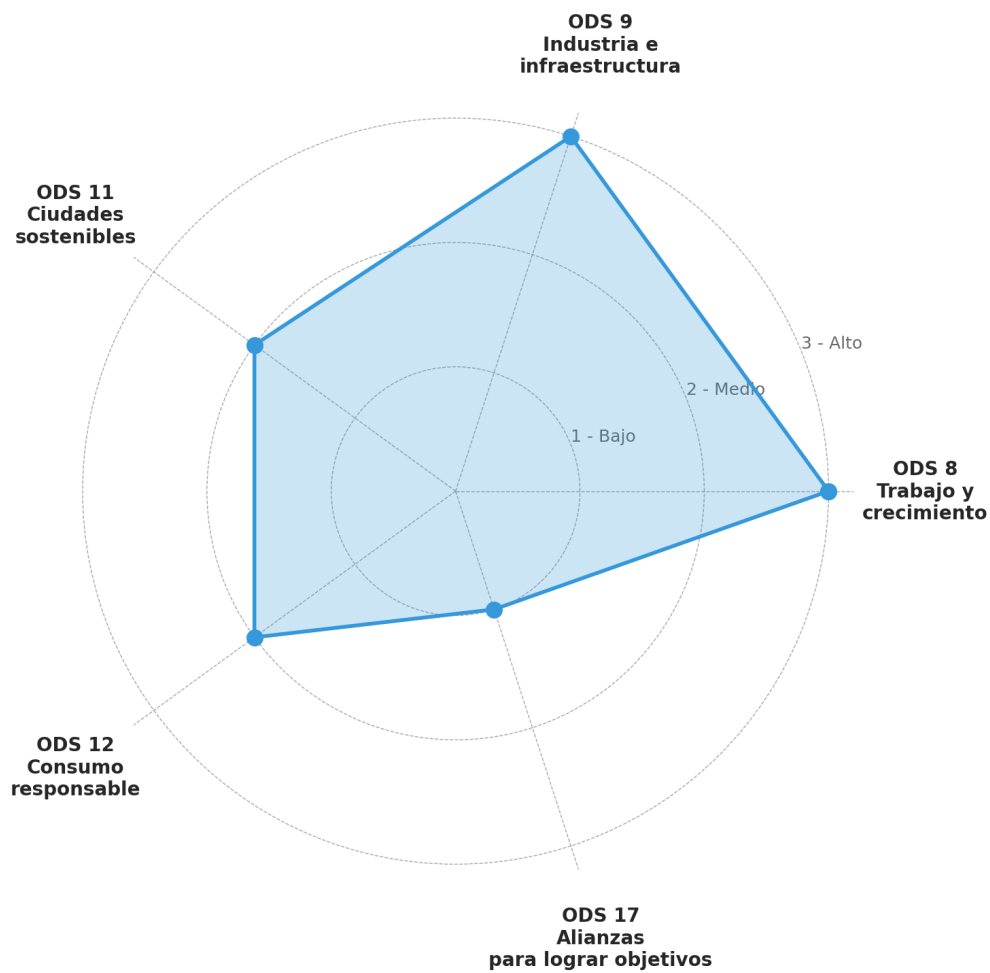
Figura 3.1: Alineamiento con los ODS

Figure 3.1: Alignment of the work with the Sustainable Development Goals.

Chapter 4

Development

The complete source code of the work, organised in ten executable and reproducible Jupyter notebooks (NB00 to NB09), is publicly available in the GitHub repository <https://github.com/DieGodMF4/Macro-Aviation-Tourism-Modeling>, together with the raw and processed datasets, the figure generation scripts and the present thesis. References to notebooks throughout this chapter and Chapter 5 use the abbreviated nomenclature NB00 to NB09, in line with the repository’s convention; each notebook has a descriptive name within the `docs/` folder that facilitates identification.

This chapter describes the methodology followed and the initial development phases of the work. In accordance with the structure agreed with the supervisor, this chapter covers the phases running from problem and data understanding to the preparation of the variable panel, namely Jupyter notebooks NB00 to NB04. The construction and evaluation of the predictive models—notebooks NB05 to NB09—are developed in Chapter 5, given that their results are inseparable from the presentation of each model. Nevertheless, this chapter introduces the common methodological framework and briefly describes the techniques to be applied subsequently, so that Chapter 5 can focus on results without needing to present the techniques from scratch.

4.1 Methodology

4.1.1 Methodological framework: CRISP-DM

The project is developed under the **CRISP-DM** (*Cross-Industry Standard Process for Data Mining*) methodological standard, widely adopted in data science practice. Its iterative, data-centred nature makes it especially suitable for econometric experimentation processes, where modelling decisions often require returning to earlier phases—for example, when exploratory analysis reveals the need to reformulate feature engineering, or when the results of one model suggest revising variable selection criteria.

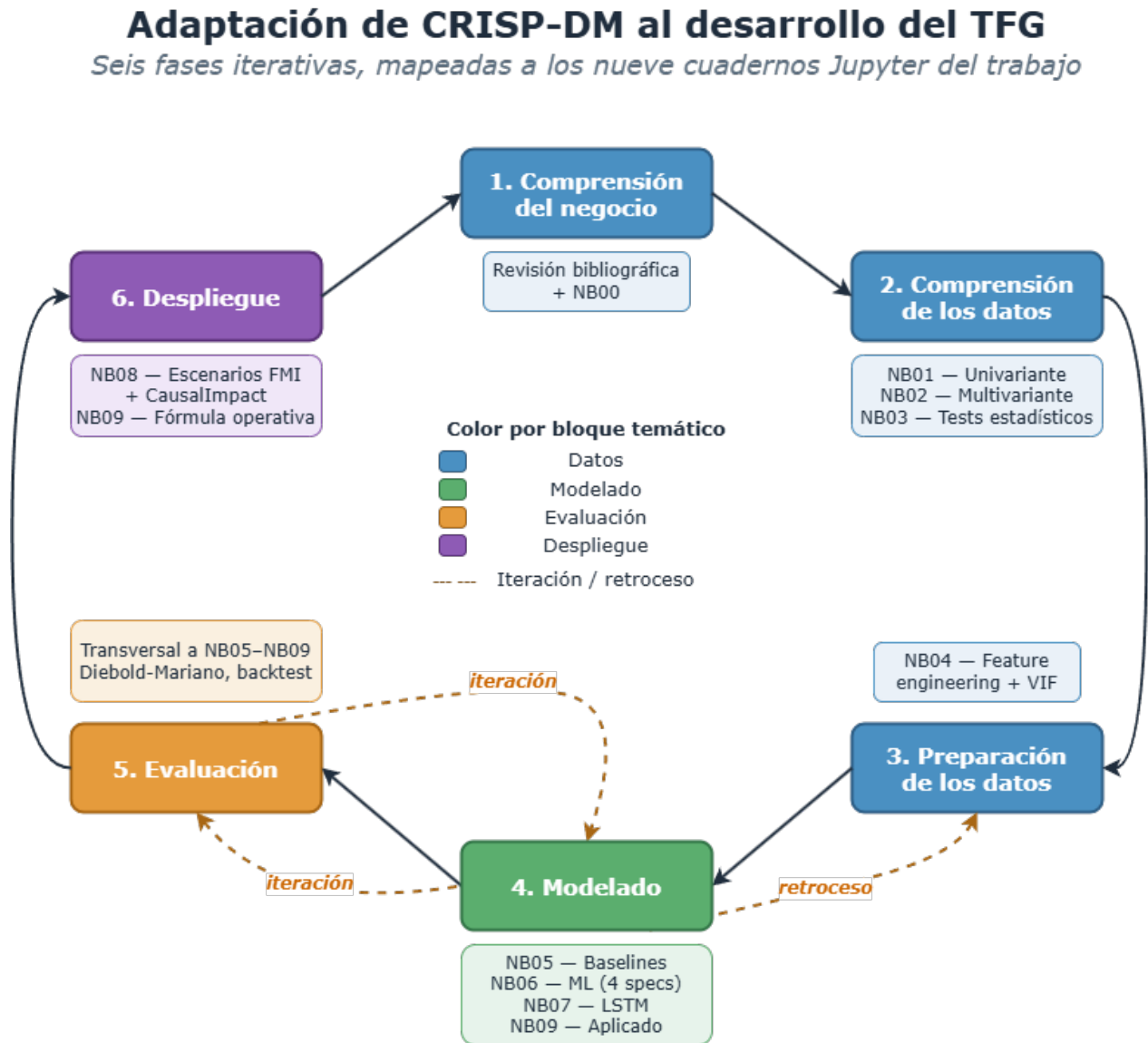


Figure 4.1: Adaptation of the CRISP-DM standard to the six phases of the work and its nine notebooks.

Figure 4.1 represents this adaptation of the CRISP-DM standard to the six phases of the work and their correspondence with the nine notebooks. The six phases of the standard are adapted to the present work as follows. **Business understanding** is covered by the literature review in Chapter 2 and the definition of objectives. **Data understanding** is implemented in notebooks NB00 to NB03, which cover structural loading and inspection, univariate and multivariate exploratory analysis, and formal statistical tests. **Data preparation** is concentrated in notebook NB04, which performs temporal interpolation of lower-frequency series, statistical chain-linking of UK series, construction of derived variables and collinearity diagnosis. **Modelling** spans notebooks NB05 (classical baselines), NB06 (machine learning), NB07 (LSTM) and NB09 (applied air capacity model). **Evaluation** is cross-cutting across all modelling notebooks and is consolidated in Chapter 5. Finally, **deployment** is materi-

alised in notebook NB08, which integrates the models into a prospective pipeline fed by IMF projections, and in the operational component of NB09.

4.1.2 Work workflow

Figure 4.2 synthesises the general workflow of the work, showing the sequence from objective definition to the communication of results.

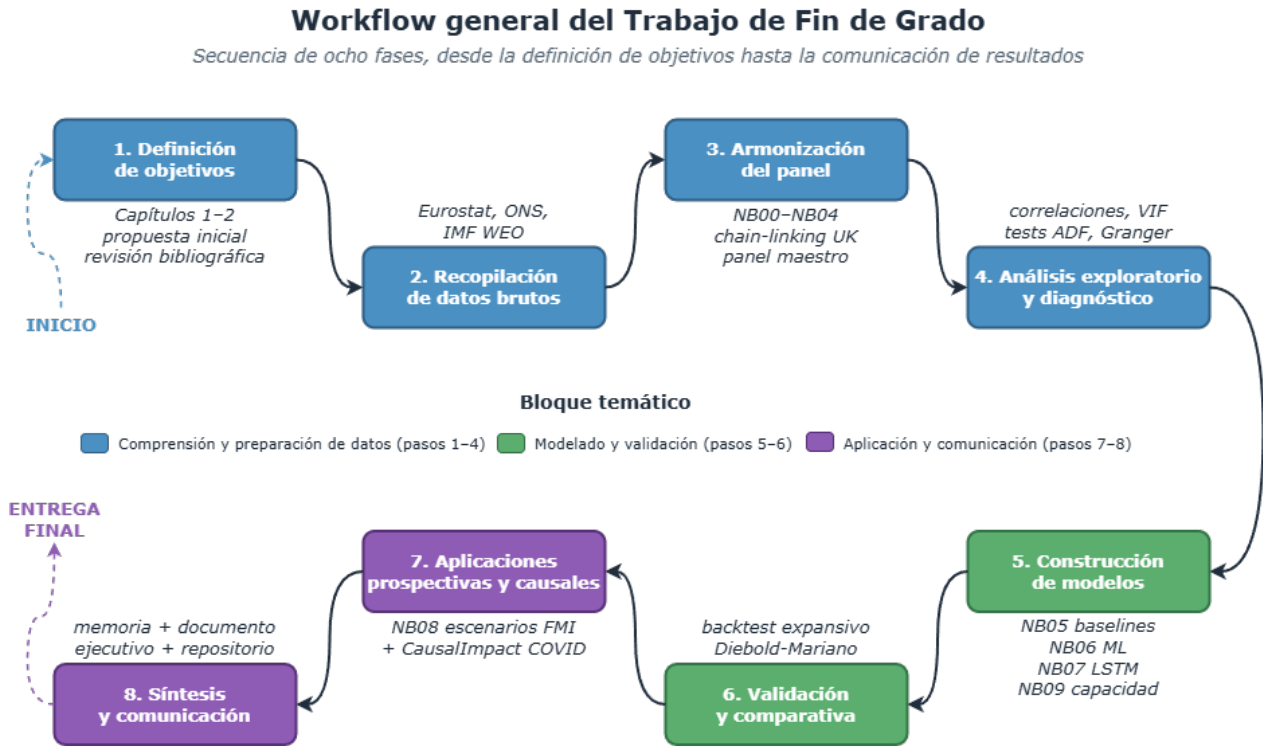


Figure 4.2: General workflow of the work in eight sequential phases.

4.1.3 Techniques used in the work

Before describing the specific data preparation phases, it is useful to briefly present the techniques to be applied throughout the work, so that the reader has an overall view of the methodological toolkit employed and the subsequent presentation of results is self-contained.

SARIMA / SARIMAX. Seasonal autoregressive integrated moving average models. SARIMA models exclusively the univariate dynamics of the target series; SARIMAX extends this structure by incorporating exogenous regressors. They are the classical reference in time series econometrics for monthly data with marked seasonality (Box and Jenkins, 1976).

Holt-Winters. Exponential smoothing with level, trend and seasonality components. Its three parameters are estimated by maximum likelihood. It represents the classical statistical state of the art for seasonal series without exogenous variables.

Seasonal Naïve. Trivial reference model that consists of forecasting the value of any future month as the value observed in the same month of the previous year: $\text{prediction}[t] =$

`observed[t-12]`. It requires no parameter estimation or fitting and serves as a minimum benchmark: any more sophisticated model must outperform the naïve to justify its use. In strongly seasonal series such as that of the present work, the seasonal naïve captures by construction all annual regularity, making it a surprisingly strong competitor at short horizons. Its performance in the baseline comparison (Section 5.1) and as a reference against the applied air capacity model (Section 5.5) will be discussed in Chapter 5.

Random Forest and XGBoost. Models based on decision tree *ensembles*. Random Forest (Breiman, 2001) averages the predictions of trees trained on *bootstrap* samples; XGBoost (Chen and Guestrin, 2016) sequentially optimises the errors of previous models via *gradient boosting*. Both capture non-linear relationships and interactions between variables without requiring prior functional specification.

LSTM (Long Short-Term Memory). Recurrent neural network designed by Hochreiter and Schmidhuber (1997) to capture long-term temporal dependencies through memory gates that control the flow of information through time, overcoming the gradient problems that affect classical recurrent architectures. Its application to tourism demand forecasting has been validated by Salamanis et al. (2022), who document significant improvements over classical models at long horizons. This work explores two architectural variants: LSTM-1step (one-step prediction, applicable recursively for longer horizons) and LSTM-direct12 (direct prediction of the full twelve-month vector in a single step).

CausalImpact and BSTS. Bayesian framework developed by Brodersen et al. (2015) for counterfactual causal inference in time series. From a *Bayesian Structural Time Series* model trained on the pre-intervention period, it constructs a counterfactual series estimating what the evolution of the target would have been in the absence of the shock. The difference between the observed and counterfactual series, integrated over time and endowed with a credibility interval, constitutes the causal impact estimate.

Evaluation metrics. Three standard metrics are used: the Root Mean Square Error (RMSE), the Mean Absolute Error (MAE) and the Mean Absolute Percentage Error (MAPE). The MAPE is the primary metric due to its interpretability, while RMSE and MAE complement the diagnostic when errors are asymmetric.

Diebold-Mariano test. Formal statistical test for comparing the predictive accuracy of two models on a common set of forecasts (Diebold and Mariano, 1995). Applied with the Harvey, Leybourne and Newbold (1997) correction to mitigate bias in small samples, a situation common in temporal validation of tourism series.

Expanding-window backtest. Temporal validation scheme in which the model is retrained at successive origins distributed across the series, evaluated at each origin over multiple horizons (1, 3, 6 and 12 months). Provides more robust precision estimates than a single hold-out and reveals the model’s sensitivity to the evaluation period.

4.2 Data sources and panel construction (NB00)

Notebook NB00 implements the loading of all raw datasets and the unification of their structure. Data come primarily from Eurostat, supplemented by series from the ONS of the United Kingdom and, for the prospective module of Chapter 5, by projections from the IMF’s *World Economic Outlook*.

The study period spans from January 2011 to November 2025, with a total of 179 monthly observations. The choice of 2011 as the starting point responds to the complete and harmonised availability of all relevant datasets following the entry into force of Regulation (EU) 692/2011 on European tourism statistics.

Target variable. Monthly overnight stays by foreign tourists in tourist accommodation establishments in Poland, extracted from Eurostat dataset `tour_occ_nim` (Eurostat, 2025a) with filters `c_resid = FOR` (residents abroad), `nace_r2 = I551-I553` (full tourist accommodation) and `unit = NR` (number). The observed range runs from 83,695 overnight stays—the minimum, recorded in April 2020 at the most severe point of lockdown—to 2,453,961—the maximum, recorded in August 2024—with a mean of around 1.19 million.

Source markets and weighting criterion. Given the structural limitation of dataset `tour_occ_nim` (Eurostat, 2025a), which does not disaggregate by country of origin, the source-market dimension is incorporated through exogenous variables weighted by the air passenger share derived from dataset `avia_paoc` (Eurostat, 2025d). The nine markets considered, with their period-average shares, are Spain (19.8%), Germany (19.5%), the United Kingdom (18.8%), France (14.4%), Italy (13.5%), the Netherlands (6.6%), Sweden (3.2%), Austria (2.9%) and the Czech Republic (1.4%). These nine countries cover approximately all air passenger traffic to Poland.

Exogenous variables and sources. Regressors are grouped into four blocks. (i) Prices: monthly Harmonised Index of Consumer Prices (`prc_hicp_midx`, base 2015=100) (Eurostat, 2025e) for Poland and the nine source markets, and bilateral exchange rates against the euro (`ert_bil_eur_m`) (Eurostat, 2025f) for non-euro currencies. (ii) Income and demographics: quarterly real GDP (`namq_10_gdp`) (Eurostat, 2025b) and annual population (`demo_pjan`) (Eurostat, 2025c), from which GDP per capita is derived. (iii) Economic sentiment: monthly Consumer Confidence Indicator (`ei_bscm`) (Eurostat, 2025g) for Germany, the source market with the largest share and the most extensive series. (iv) Air connectivity: monthly available seats at Polish airports (`avia_tf_aca`) (Eurostat, 2025h).

Integration of the United Kingdom via Chain-linking (statistical chain-linking).

Following the United Kingdom’s withdrawal from the European Statistical System, its series in Eurostat are interrupted or present methodological breaks that are difficult to compare with those of the remaining markets. Since the United Kingdom accounts for the third largest share of air traffic to Poland, excluding or extrapolating it would be seriously problematic. The adopted solution is statistical chain-linking with official ONS series: for each affected series—HICP (ONS series D7BT), real GDP (ABMI) and population (UKPOP)—an overlap period with the Eurostat series is identified, an adjustment factor is calculated that equalises the levels at the splicing point, and a composite series is constructed that uses Eurostat before

the splice and ONS (rescaled) afterwards. This procedure, standard in applied econometrics, preserves the effective growth rates from the ONS while maintaining level harmonisation with the rest of the panel. Figure 4.3 shows the result of the statistical chain-linking applied to the UK's HICP and GDP: the Eurostat and ONS segments join smoothly at the splicing point, evidencing that the rescaling factor has been correctly estimated.

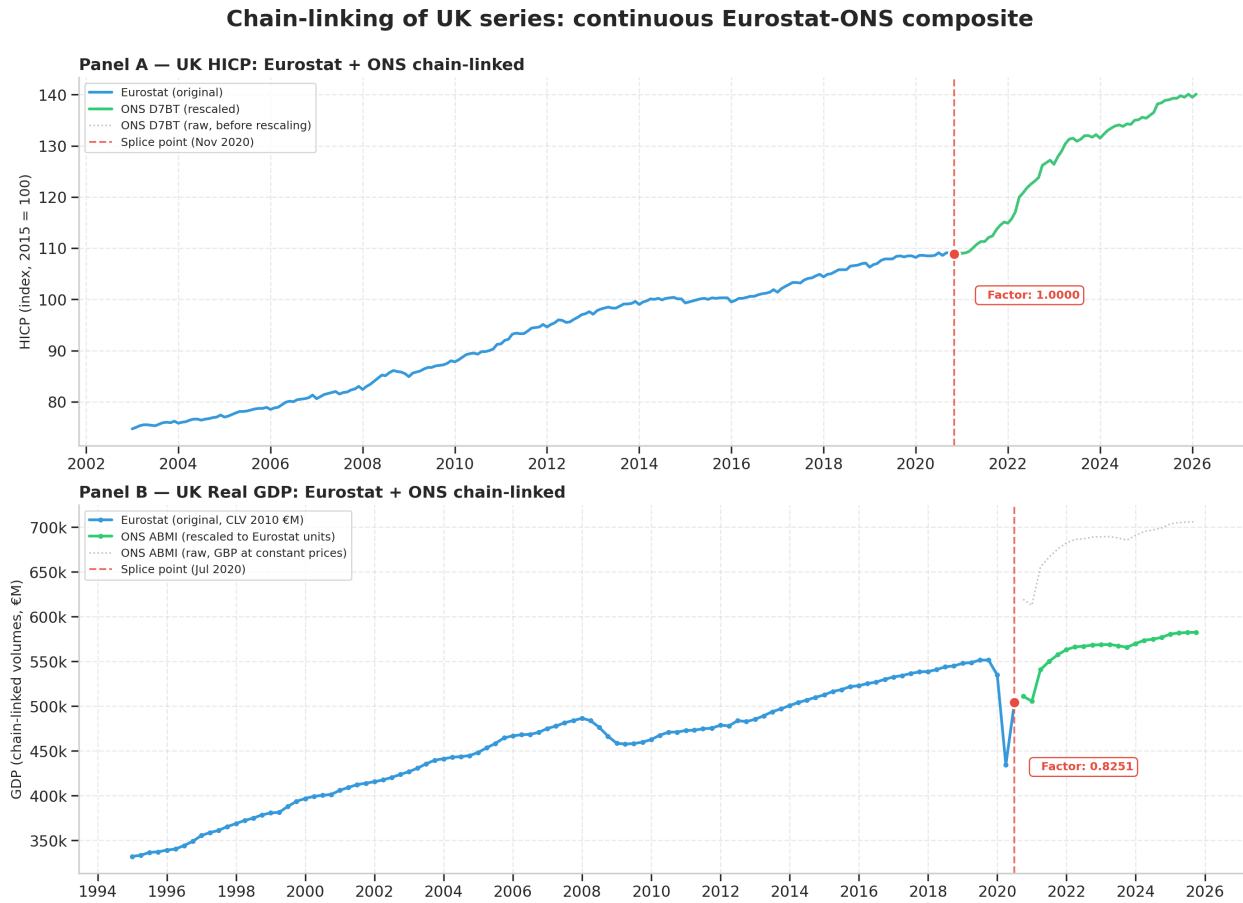


Figure 4.3: Chain-linking of UK series: HICP (Panel A) and real GDP (Panel B).

4.3 Univariate exploratory analysis (NB01)

Notebook NB01 characterises the target variable through time visualisation, seasonal decomposition and distributional analysis. The most relevant structural features are a **strongly marked seasonality**, with a ratio between the August peak (around 1.87 million in the pre-pandemic period) and the January trough (around 783,000) close to 2.4; a **severe COVID-19 impact**, with a year-on-year collapse of 96.3% in April 2020 and a recovery period of approximately 28 months; and a **share of foreign tourists over the total** of around 18.8% on average, placing Poland as a predominantly domestic tourism destination, though with an international segment growing at a faster pace and constituting the focus of the present work. These features are represented in Figure 4.4: Panel A shows the monthly series with its smoothed trend and the COVID shock shaded, Panel B represents year-on-year growth

rates with the 2020–2021 collapse clearly visible, and Panel C presents the year-by-month heat map that highlights the summer concentration and the pandemic “dark patch”.

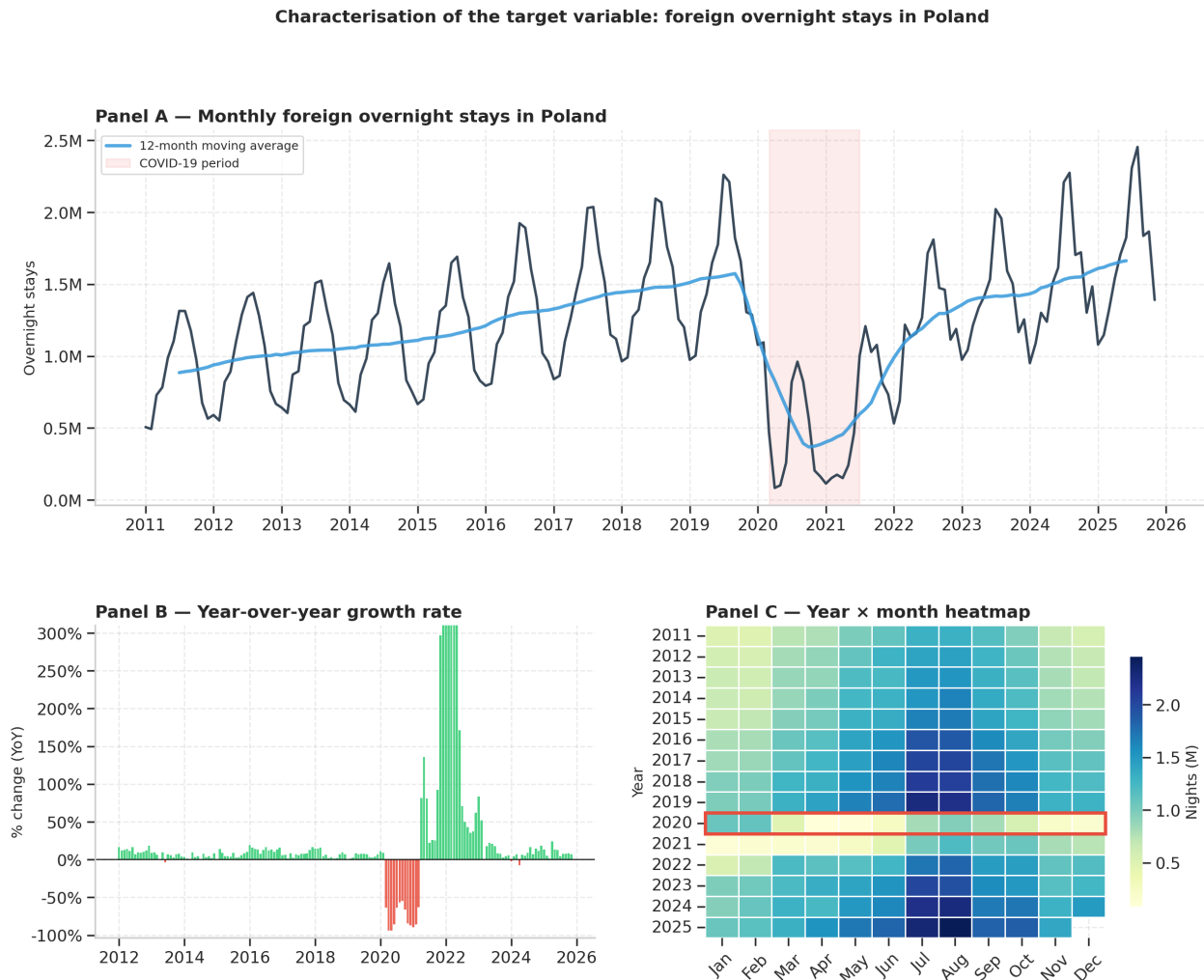


Figure 4.4: Characterisation of the target variable: monthly series (A), year-on-year growth (B) and year $\times$ month heat map (C).

As a complement to the above descriptive analysis, Figure 4.5 presents the STL (*Seasonal-Trend decomposition using Loess*, Cleveland et al. 1990) decomposition of the target series into its three canonical components: trend, annual seasonality and residual. Three observations merit comment.

The **underlying trend** grows steadily from 2011 until the pandemic shock, is abruptly interrupted in 2020 and resumes an even steeper positive slope from 2022, settling at post-COVID levels approximately 40 % above pre-COVID ones. This structural level shift is the phenomenon that the year-on-year LSTM formulation (Chapter 5) is designed to absorb automatically.

The **seasonal component** exhibits a notably stable amplitude across the fourteen years of

the sample, with a recurring peak in July–August and a trough in January. This stability—confirmed quantitatively in Chapter 5 with correlations between pre- and post-pandemic seasonal patterns above 0.94—is the empirical property that justifies the use of a single seasonal model for the entire period.

The **residual** clearly concentrates the effect of COVID-19 between 2020 and 2022, at which point it exceeds in magnitude the variability characteristic of the rest of the series. Outside that period, the residual behaves approximately as white noise, indicating that the first two components (trend + seasonality) capture virtually all systematic structure in the series.

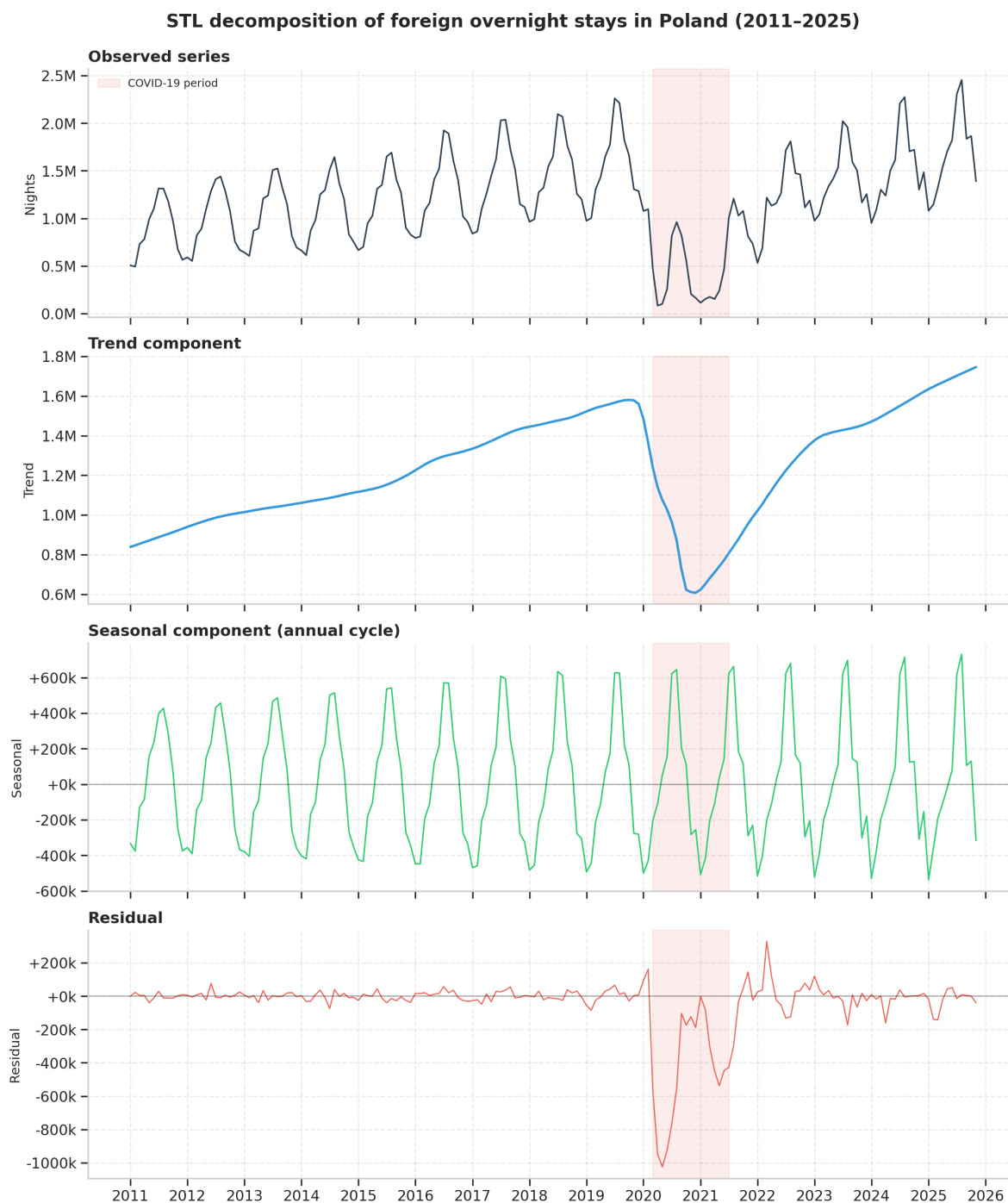


Figure 4.5: STL decomposition of foreign overnight stays in Poland (2011–2025).

4.4 Multivariate exploratory analysis and statistical diagnostics (NB02–NB03)

Notebook NB02 examines the relationships between the target variable and candidate regressors through correlation matrices and cross-correlation functions (CCF) at lags 0 to 12 months, contrasting results obtained in absolute levels against those obtained in year-on-year (YoY) growth rates. The contrast between both representations is extraordinarily revealing and guides the remaining methodological decisions of the work. Figure 4.6 presents both heat maps side by side to facilitate comparison between the trend-contaminated correlations (Panel A) and the clean year-on-year correlations (Panel B).

Cross-correlation between nights(t) and predictor(t-k)

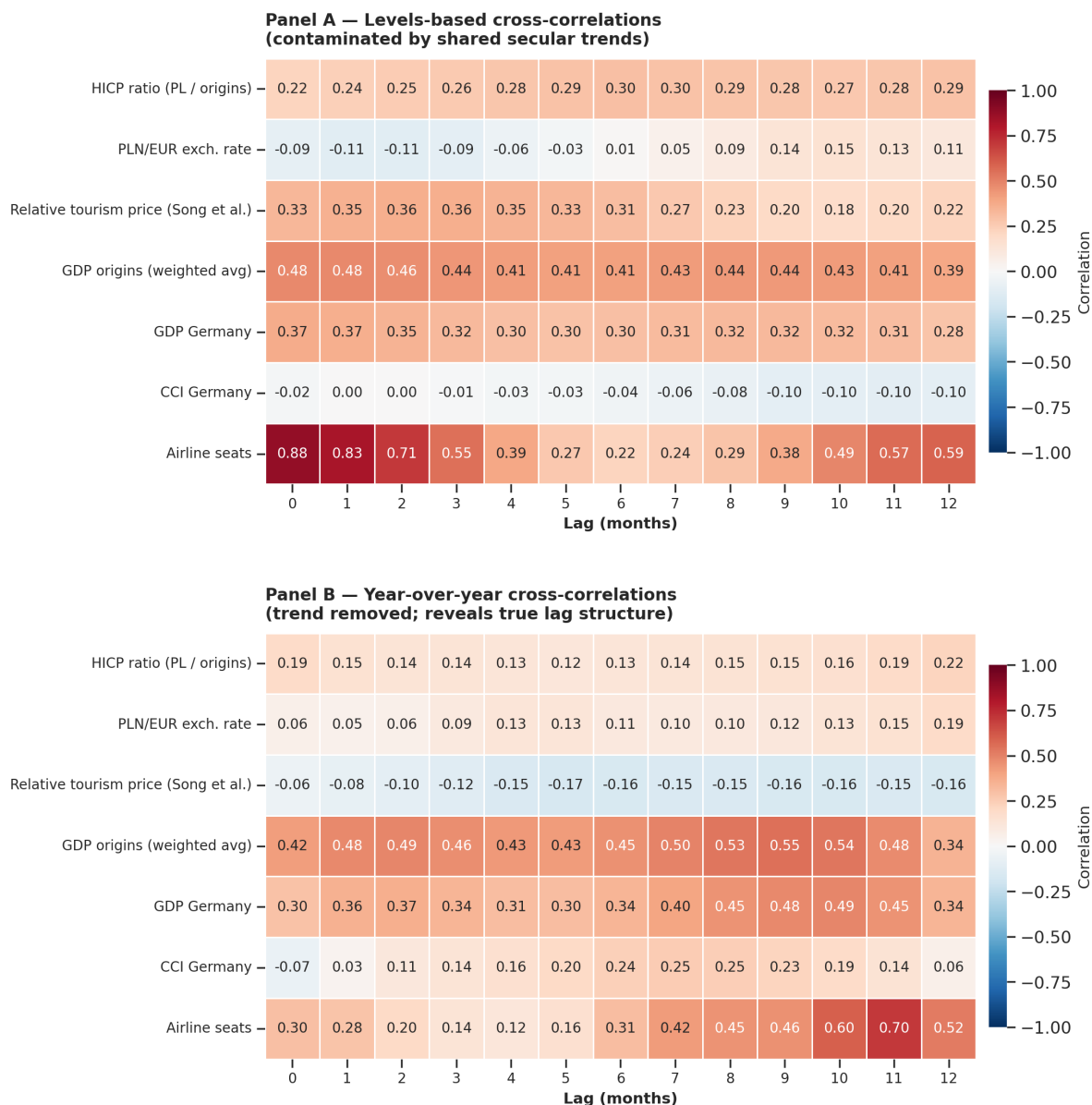


Figure 4.6: Cross-correlations between the target variable and the main regressors at lags 0–12 months, in levels (Panel A) and in year-on-year growth rates (Panel B).

Correlations in levels. Contemporaneous air capacity (**seats**) presents the highest correlation with the target ($r = 0.88$ at lag 0), followed by the weighted source-market GDP ($r = 0.48$ at lag 0) and German GDP ($r = 0.37$ at lag 0). The remaining macroeconomic regressors—HICP ratio, exchange rate, relative tourist price, consumer confidence—present low or very low correlations in absolute value (all below 0.40). These level values **are contaminated by the shared secular trend**: all macroeconomic series exhibit a sustained

upward trend from 2011 that produces spurious correlations with tourism demand, which is also trending upward. Therefore, substantive inference must be based on the year-on-year analysis.

Correlations in year-on-year growth rates. After removing the trend component, the ranking changes substantially and, most importantly, a non-trivial structure of optimal lags emerges:

- ✓ **Air capacity** remains the dominant predictor, but with a radically different temporal pattern: the contemporaneous correlation falls to $r = 0.30$, while the peak shifts to **lag eleven** ($r = 0.70$). This is the clearest empirical evidence in favour of the methodological decision to use `seats_lag6` or long lags rather than contemporaneous capacity: capacity scheduling decisions anticipate demand by many months, in line with the standard IATA Slots allocation horizon.
- ✓ The **weighted source-market GDP** maintains a robust and stable correlation of around $r = 0.42$ – 0.55 across all lags from 0 to 11, peaking at lag 9 ($r = 0.55$). The same pattern is observed, with slightly smaller magnitudes, for German GDP (peak at lag 10 with $r = 0.49$).
- ✓ The **German consumer confidence indicator** confirms its theoretical role as a leading indicator: the contemporaneous correlation is practically zero ($r = -0.07$) but grows progressively to reach its peak around lags 7–8 ($r = 0.24$ – 0.25), consistent with the intuition that households decide and plan their trips several months before they take place.
- ✓ **Price and exchange rate variables** (HICP ratio, PLN/EUR, relative tourist price) present weak linear correlations in year-on-year growth rates (all with $|r| \leq 0.22$). This finding does not imply that they are irrelevant to demand, but rather that their relationship with the target is not well captured by a monotone linear correlation. This is one of the empirical motivations that justify the shift from classical econometric models (linear by construction) to machine learning and deep learning models, capable of capturing non-linear relationships and interactions between variables.

Notebook NB03 formalises this exploratory analysis with statistical tests of the properties of the series. ADF and KPSS tests confirm that the target series is non-stationary in levels and becomes stationary upon taking seasonal differences. Granger causality tests detect temporal precedence between the majority of macroeconomic regressors and the target variable, consistent with the theoretically expected direction. The most relevant finding of this notebook, however, is the **detection of Granger causality in both directions between air capacity and tourism demand**, empirically confirming the suspicion of bidirectional Endogeneity already noted in Chapter 2 and motivating the four-specification scheme of NB06. Figure 4.7 summarises the complete results in a bidirectional matrix of p-values.

**Granger causality tests:
minimum p-value across lags 1-6**

Airline seats	<0.001 ***	<0.001 ***
GDP origins (weighted avg)	<0.001 ***	0.096
GDP Germany	<0.001 ***	0.124
HICP ratio (PL / origins)	<0.001 ***	<0.001 ***
PLN/EUR exchange rate	0.069	0.252
Relative tourism price (Song et al.)	0.109	0.043 *
CCI Germany	0.009 **	0.391
	X → nights	nights → X

Significance: * $p < 0.05$ ** $p < 0.01$ *** $p < 0.001$
Green cells indicate significant Granger causality ($p < 0.05$).

Figure 4.7: Matrix of p-values of Granger causality tests between the target variable and the main regressors, in both directions.

Three qualitative findings emerge from the analysis. First, the macroeconomic regressors that are genuinely exogenous to the system—weighted source-market GDP, German GDP and the German CCI—exhibit significant Granger causality exclusively in the theoretically expected direction ($X \rightarrow \text{nights}$), with no statistical reciprocity in the reverse direction. This is consistent with economic intuition: the macroeconomic conditions of source markets precede Polish tourism demand, but Polish tourism flows do not have the capacity to move German GDP.

Second, and central to the methodological strategy of the work, **air capacity (seats) exhibits highly significant bidirectional Granger causality ($p < 0.001$ in both directions)**, empirically confirming the suspicion of bidirectional endogeneity already noted in Chapter 2. This result motivates the four-specification scheme of NB06 developed in Chapter 5 and justifies the exclusion of contemporaneous capacity as a predictor in favour of the `seats_lag6` lag predetermined by the IATA horizon.

Third, the HICP ratio between Poland and its source markets also shows significant bidirectional causality, suggesting that relative inflation not only influences tourism flows but is also modulated by them through the pressure that tourism exerts on domestic prices. Although this finding does not affect the modelling strategy—the HICP ratio is incorporated with its contemporaneous value in the panel—it is worth noting as a descriptive finding.

4.5 Feature engineering and collinearity diagnostics (NB04)

Notebook NB04 completes the data preparation phase with the construction of derived variables, the treatment of COVID-19 via dummy variables and a systematic collinearity diagnostic.

Derived variables. The relative tourist price (`relprice_wavg`) is constructed according to the Song et al. (2010) formulation described in Chapter 2, the weighted source-market GDP per capita (`gdp_pc_origins_wavg`) and the weighted population (`pop_origins_wavg`). Following an explicit recommendation from the supervisor—subsequently validated by VIF—the pair **GDP per capita plus population** is adopted as complete information, rather than the trio aggregate GDP plus GDP per capita plus population, which would be mathematically redundant.

COVID-19 treatment. Two complementary dummies are constructed: `covid_strong` (equal to 1 between March 2020 and December 2021), capturing the acute phase of the collapse, and `covid_post` (equal to 1 from January 2022), capturing the permanent post-pandemic level shift. This two-phase structure allows the model to simultaneously estimate the effect of the acute collapse and the subsequent level displacement, avoiding the need for manual corrections.

Lagged variables. In time series analysis, a lagged variable `lag_k` represents the value of that variable observed k months before the reference period: thus, `nights_lag12` denotes the overnight stays that occurred twelve months before the current month, and `cci_DE_lag2` denotes the German Consumer Confidence Indicator observed two months earlier. Incorporating lagged variables into a predictive model serves three complementary purposes. First, it captures the **temporal inertia** of the series: tourism demand in a given month is strongly correlated with that of the same month the previous year for evident seasonal reasons, and this correlation is exploitable information materialised in the twelve-month lag. Second, it allows the incorporation of explanatory variables that act with a **temporal delay**, such as leading consumer confidence indicators—which typically anticipate spending decisions by one or two months. Third, and particularly relevant for this work, lags allow **real operational constraints** to be reflected: airlines plan capacity several months in advance in line with

IATA guidelines on airport Slots allocation, which justifies the use of air capacity observed six months earlier (`seats_lag6`) as a predictor of contemporaneous demand without incurring the bidirectional Endogeneity problem documented in Chapter 2.

With these three justifications, the lagged variables constructed in this notebook are `nights_lag1` and `nights_lag12` (autoregressive lags of one and twelve months of the target variable itself), `gdp_origins_lag3` (aligned with the cross-correlation peak of the source-market GDP observed in NB02), `cci_DE_lag2` (aligned with the leading dynamics of the German consumer confidence indicator) and `seats_lag6` (predetermined by the standard IATA slot allocation horizon). The choice of each specific lag is empirically documented: the lag for each variable is identified as that which maximises the cross-correlation with the target variable within a temporally reasonable range, thus avoiding the arbitrary introduction of lagged variables without justification.

Collinearity diagnostics. The Variance Inflation Factor (VIF) is calculated on first differences, which is the relevant metric when series share a secular trend and the level version would produce artificially inflated values. The diagnostic identifies two groups of variables with collinearity problems:

- ✓ **Severe collinearity ($VIF > 20$):** `gdp_pc_origins_wavg` ($VIF = 23.97$) and `gdp_pc_DE` ($VIF = 22.86$) exhibit significant mutual collinearity, a consequence of Germany's approximately 20% weight in the weighted aggregate, meaning both variables share a substantial portion of their variation. The direct bivariate correlation between them is 0.95.
- ✓ **Moderate collinearity ($5 < VIF < 10$):** `relprice_wavg` ($VIF = 9.14$) and `exr_pln` ($VIF = 8.97$) exhibit moderate collinearity with the rest of the price and exchange rate block, arising because both share dynamics with the HICP ratio through the bilateral exchange rate.

These findings, together with the bidirectional Endogeneity of air capacity documented in NB03, are synthesised in a **reproducible decision table** (Figure 4.8) that assigns one of three actions to each candidate variable. DROP actions due to Endogeneity affect contemporaneous `seats` and `nights_lag1`: both are highly predictive ($r = 0.88$ with the target) but their use as predictors would produce academically informative and operationally unrealisable models, in line with the distinction articulated in Section 4.6. CHECK actions affect the four variables with $VIF > 5$: they are not eliminated a priori, but rather evaluated empirically by specification in NB06 through the four-model scheme of decreasing complexity (M1 to M4), which quantifies how much predictive accuracy is lost when collinear information is progressively restricted. KEEP actions affect the remaining variables in the feature set, including non-endogenous serial lags (`nights_lag12`, `cci_DE_lag2`, `seats_lag6`) and the two COVID dummies.

Variable selection decision table (NB04)

Variable	r (target)	VIF (Δ)	Endogenous	Collinear	Action
hicp_ratio	+0.096	1.87	No	No	KEEP
relprice_wavg	+0.249	9.14	No	Yes	CHECK (VIF>5)
exr_pln	-0.135	8.97	No	Yes	CHECK (VIF>5)
gdp_pc_origins_wavg	+0.444	23.97	No	Yes	CHECK (VIF>5)
pop_origins_wavg	+0.195	1.15	No	No	KEEP
gdp_pc_DE	+0.342	22.86	No	Yes	CHECK (VIF>5)
cci_DE	+0.057	1.38	No	No	KEEP
seats	+0.877	2.21	Yes	No	DROP (endogenous)
nights_lag1	+0.877	1.44	Yes	No	DROP (endogenous)
nights_lag12	+0.629	1.58	No	No	KEEP
cci_DE_lag2	+0.068	1.09	No	No	KEEP
seats_lag6	+0.167	1.48	No	No	KEEP
covid_strong	-0.540	—	No	No	KEEP
covid_post	+0.243	—	No	No	KEEP

KEEP — variable retained CHECK — collinear, evaluated by specification DROP — endogenous or redundant

Figure 4.8: Decision table for variable selection.

As an illustrative example of the coding style used throughout the work, the snippet from NB04 that implements the statistical chain-linking of the UK HICP is reproduced below; it is one of the most technically delicate operations in the data pipeline. The complete code is available in the project repository.

```
def chain_link(eurostat_series, ons_series, overlap_start, overlap_end):
    """Links two time series over the overlap period,
    preserving the growth rates of the ONS series and rescaling
    to the Eurostat level at the splicing point.
    """
    overlap_es = eurostat_series.loc[overlap_start:overlap_end]
    overlap_on = ons_series.loc[overlap_start:overlap_end]
    # Adjustment factor via least squares over the overlap period
    scale_factor = (overlap_es * overlap_on).sum() / (overlap_on ** 2).sum()
    # Construction of the composite series
    composite = pd.concat([
        eurostat_series.loc[:overlap_start - pd.DateOffset(days=1)],
        ons_series.loc[overlap_start:] * scale_factor,
    ])
    return composite.sort_index()
```

Algorithm 4.1: Statistical chain-linking function implemented in NB04.

4.6 Academic versus operational distinction in variable selection

A methodological decision running throughout the rest of the work, articulated in dialogue with the supervisor, is the explicit distinction between the academic objective of inferring the determinants of demand and the operational objective of producing useful forecasts under constraints on the temporal availability of information. The difference is not theoretical: in an academic context it is correct to retain short-term autoregressive lags (`nights_lag1`) because first-order autocorrelation is well documented and allows the inertia of the series

to be correctly modelled; in an operational context, an airline planning capacity six months ahead does not have the value of `nights_lag1` when making the decision, and therefore a model trained and evaluated using `nights_lag1` produces academically informative but operationally unrealisable forecasts.

This distinction justifies the systematic evaluation in Chapter 5 of several specifications of the same model, ranging from the most complete to one restricted to operationally available information, explicitly measuring the cost of operational honesty. The quantified difference between both specifications is one of the relevant findings of the work and will be reported in the following chapter.

4.7 Summary of the state at the close of Chapter 4

At the close of Chapter 4, the work has at its disposal a consolidated monthly panel of 179 observations $\times$ more than 30 variables, with UK series chain-linked to Eurostat, COVID dummies defined, lagged variables constructed, redundancies diagnosed and eliminated, and the academic-operational distinction explicitly formulated. On this panel, the predictive models are built in Chapter 5 and the results that materialise the objectives of the work are obtained. For a detailed consultation of these variables with their purpose and meaning, see Table 6.1 in Annex A.

Chapter 5

Results

This chapter presents and discusses the results of the five modelling notebooks of the work: NB05 (classical baselines), NB06 (machine learning), NB07 (LSTM), NB08 (prospective scenarios and COVID causal analysis) and NB09 (applied air capacity model). The order of presentation deliberately follows increasing complexity, so that each model is evaluated against its predecessors and findings accumulate progressively.

5.1 Classical baselines (NB05)

Notebook NB05 implements and evaluates four baselines: Seasonal Naïve, Holt-Winters, SARIMA and SARIMAX, compared over the last twelve months of the panel as the Hold-out.

Results are summarised in Table 5.1.

Table 5.1: MAPE of classical baselines on the 12-month hold-out.

Model	RMSE	MAE	MAPE
Holt-Winters	92,651	64,078	3.9 %
SARIMA(1,1,1)(1,1,1) ₁₂	128,826	106,085	6.0 %
SARIMAX(1,1,1)(1,1,1) ₁₂	138,428	118,505	6.6 %
Seasonal Naïve	167,404	149,398	9.1 %

Figure 5.1 visualises these forecasts on the hold-out, where it can be seen that all four baselines correctly capture the seasonal shape but differ in their ability to track the dynamics of the last year.

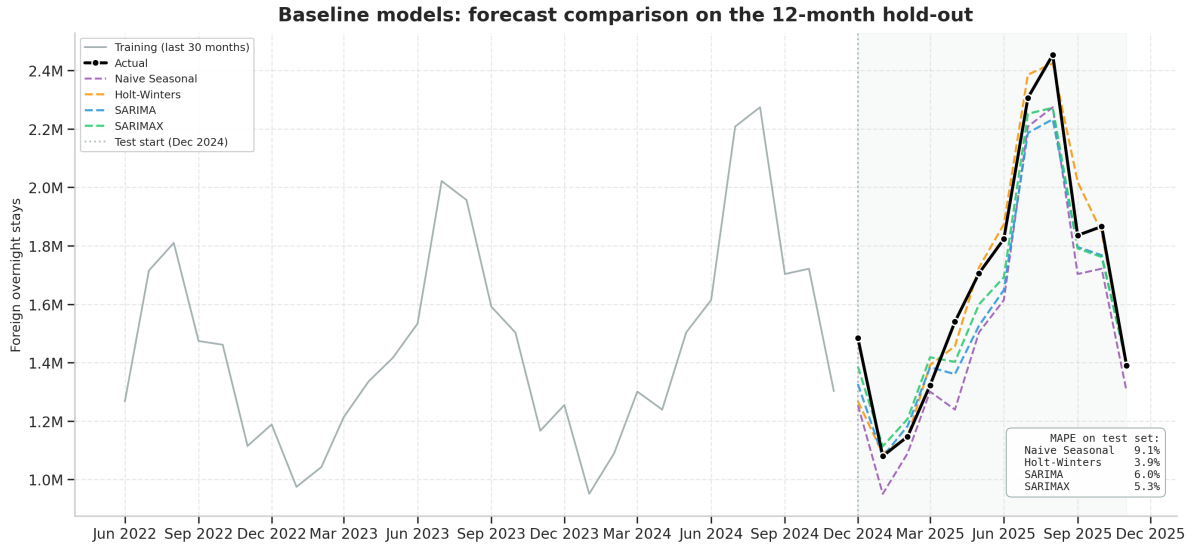


Figure 5.1: Classical baseline forecasts on the 12-month hold-out.

Two findings merit detailed commentary.

The apparent “success” of Holt-Winters is a numerically degenerate solution. Its MAPE of 3.9% outperforms the other models by a wide margin, but inspection of the optimised parameters reveals a degenerate configuration: $\alpha = 1$ (the model practically ignores the level moving average and uses the last observed value), $\beta = 0$ (no trend component) and $\gamma = 0$ (no seasonal component). This configuration is equivalent to a model that is highly reactive to the last available data point, producing accurate forecasts in stable periods but collapsing entirely in the face of structural changes. It does not constitute a robust result and is reported with that explicit caveat.

SARIMAX marginally underperforms the univariate SARIMA. This finding is counterintuitive but empirically robust: incorporating exogenous variables into SARIMA does not improve precision but slightly worsens it (MAPE 6.6% vs 6.0%). The interpretation, consistent with Gunter and Smeral (2016) and validated by the supervisor, is that in linear models the autoregressive lags already absorb the systematic variation that the exogenous variables might otherwise explain. Adding them in a saturated linear structure introduces estimation noise without informational gain. This finding conceptually motivates the move to non-linear models in the subsequent notebooks.

5.2 Machine learning models with four specifications (NB06)

Notebook NB06 implements Random Forest and XGBoost under four comparative specifications, designed to empirically isolate the marginal contribution of each informational block and resolve the air capacity Endogeneity problem described in previous chapters.

The four specifications, in order of decreasing complexity:

- ✓ **M1-Full:** all available features, including contemporaneous air capacity and short autoregressive lags. This is the “naive” specification that an analyst would produce without diagnosing endogeneity.
- ✓ **M2-Curated:** M1 minus identified redundancies and minus contemporaneous capacity (removed due to the bidirectional endogeneity confirmed in NB03). Retains `nights_lag1` and `nights_lag12`.
- ✓ **M3-NoLag1:** M2 minus `nights_lag1`. Retains `nights_lag12` (seasonal memory) and `seats_lag6` (predetermined by the IATA horizon). This is the **honest operational specification**: it uses only information available at the time of the decision.
- ✓ **M4-MacroOnly:** M3 minus serial lags. Only macroeconomic and control features. Extreme stress test.

The results of the expanding-window backtest on the ex-COVID period, at a 12-month horizon, are presented in Table 5.2. The columns of that table indicate the number of months in advance at which the forecast is made.

Table 5.2: MAPE in ex-COVID expanding backtest by specification and forecast horizon.

Model and specification	1 month	3 months	6 months	12 months
RF M1-Full	6.2 %	7.0 %	7.2 %	8.2 %
RF M2-Curated	6.2 %	7.1 %	7.2 %	8.7 %
RF M3-NoLag1	7.6 %	10.3 %	12.8 %	21.2 %
RF M4-MacroOnly	19.4 %	24.0 %	23.4 %	25.4 %
XGB M1-Full	5.4 %	6.6 %	6.9 %	7.6 %
XGB M2-Curated	5.9 %	7.5 %	8.3 %	9.4 %
XGB M3-NoLag1	7.1 %	10.3 %	12.6 %	14.3 %
XGB M4-MacroOnly	11.2 %	17.5 %	19.6 %	16.9 %

Figure 5.2 visually represents the MAPEs from Table 5.2 against the forecast horizon, separating Random Forest (left panel) and XGBoost (right panel).

Reading the findings. The progression $M1 \rightarrow M2 \rightarrow M3 \rightarrow M4$ quantifies the empirical cost of each methodological restriction.

The step from M1 to M2 removes contemporaneous (endogenous) air capacity and redundancies. The precision drop is small, demonstrating that contemporaneous capacity was contributing less predictive value than its gross importance suggested: it was endogenous and dominant in attribution but its net predictive power was moderate.

The step from M2 to M3 removes the short-term autoregressive lag. This is the largest drop, particularly in Random Forest (12.5 percentage points) and also significant in XGBoost (4.9 points). This is, literally, **the operational cost of honesty**: what is lost by not being able to use `nights_lag1` because it will not be available at the time of the decision.

The step from M3 to M4 removes the remaining serial lags (`nights_lag12` and `seats_lag6`). The additional drop indicates that these lags provide genuine information beyond purely macroeconomic features, validating their inclusion in the operational feature set.

Redistribution of variable importance. When autoregressive shortcuts are progressively removed, importance redistributes healthily towards macroeconomic variables. In M1, three variables (contemporaneous capacity, `nights_lag1` and `nights_lag12`) concentrate 97 % of the importance, leaving macroeconomic variables with practically no contribution. In M3, by contrast, `cci_DE_lag2`, `hicp_ratio` and `exr_pln` emerge with visible importance, empirically validating the theoretical determinants identified in the literature once the shortcuts that masked them are removed. Figure 5.3 shows this progression visually: cells marked with a dash indicate that the variable was not part of the feature set for that specification.

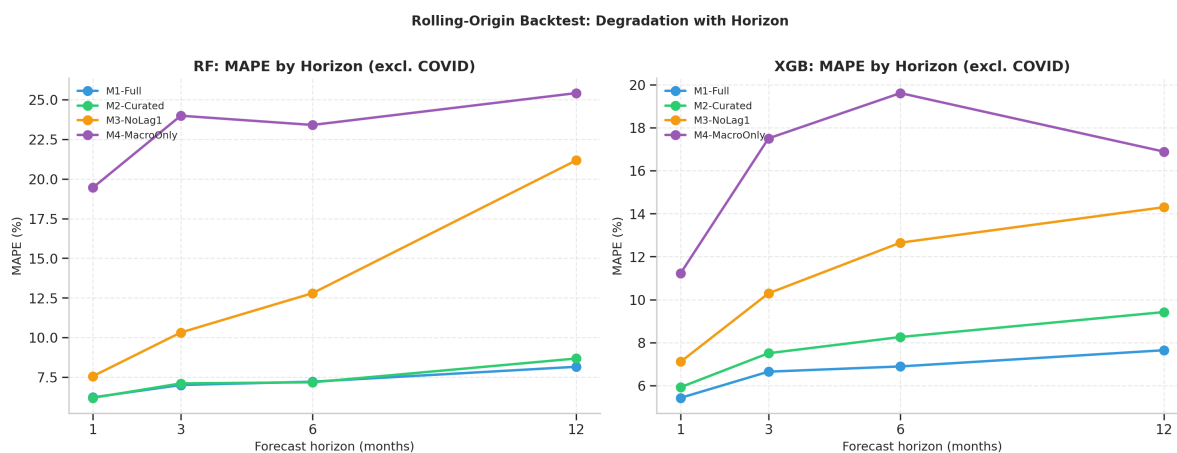


Figure 5.2: MAPE in ex-COVID expanding backtest by horizon and specification, Random Forest (left) and XGBoost (right).

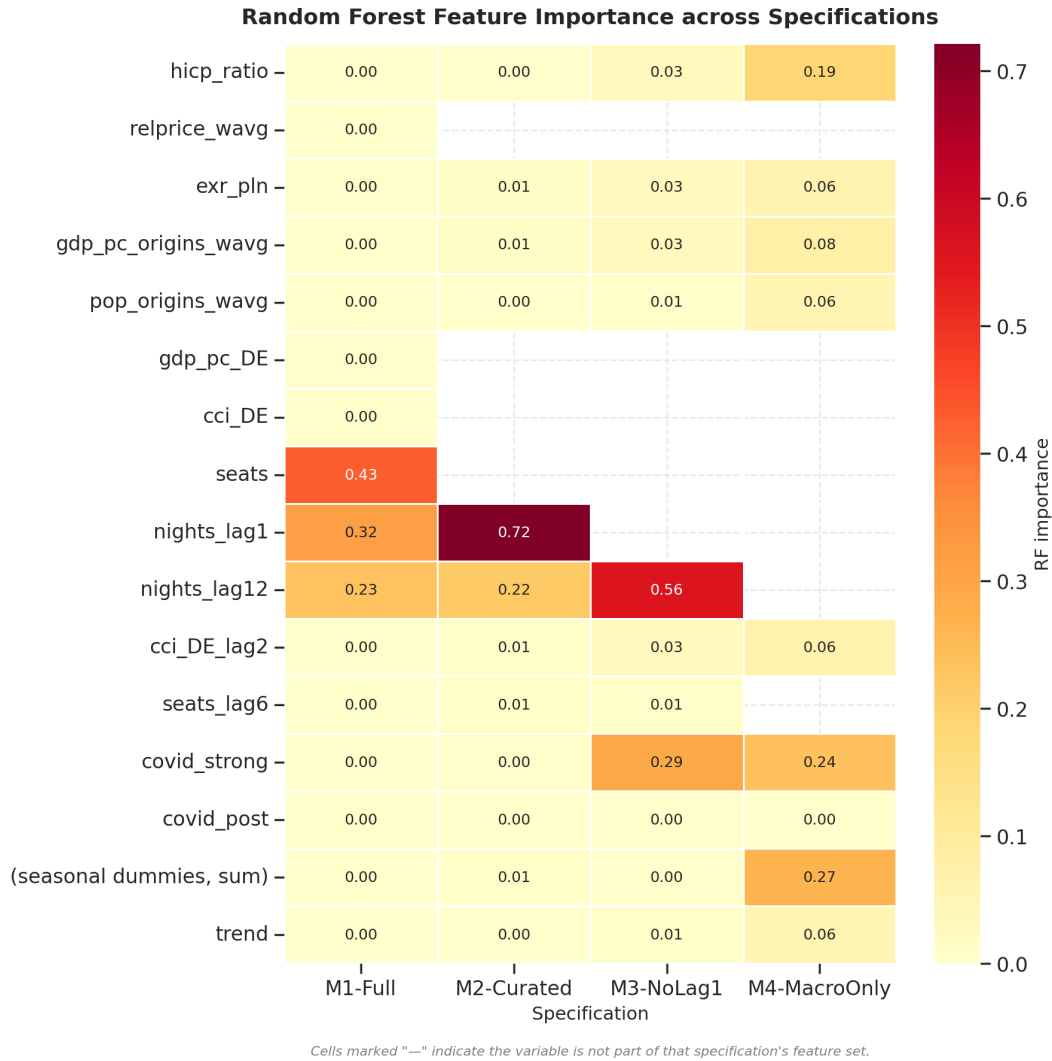


Figure 5.3: Random Forest variable importance across the four specifications M1–M4.

5.3 Deep learning with year-on-year transformation (NB07)

Notebook NB07 implements two LSTM architectures (1step recursive and direct12) under two target formulations (levels and year-on-year ratio), for a total of four variants. The motivation for the year-on-year formulation is developed below; results are reported afterwards.

5.3.1 Diagnosis of three regimes and motivation for the year-on-year formulation

Before training the models, notebook NB07 carries out an empirical diagnosis of the target series, decomposing it into three regimes: pre-COVID (2011–2019), recovery (2022–2023) and stable post-COVID (2024–2025). The analysis reveals that **correlations between the seasonal patterns of the three regimes** are extraordinarily high (between 0.94

and 0.98), indicating that the seasonal shape of the year in Poland is essentially the same across the pandemic shock. In contrast, **absolute levels change substantially**, with the post-COVID average approximately 40 % above the pre-COVID one. Year-on-year growth rates, for their part, are comparable across regimes (means between 7.7 % and 9.7 %, similar standard deviations).

These observations have a direct methodological implication. An LSTM that learns absolute levels directly has to simultaneously learn the seasonal shape (which is stable) and the level (which is not). The level component dominates the loss function and reduces capacity to learn the shape, producing a model that fails when the evaluation regime differs from the dominant training regime. Reformulating the target as the year-on-year ratio $\text{night}[t]/\text{night}[t-12]$ neutralises the level shift—which remains implicit in the denominator, observed and month-specific—and allows the network to learn only the stable component.

5.3.2 Training protocol and COVID exclusion

Recurrent networks are sensitive to structural discontinuities: with approximately 25 COVID-affected observations in a set of around 150 sequences, they do not have sufficient sample to learn the conditional logic implicit in a dummy variable. The LSTM therefore explicitly excludes from training the period between January 2020 and December 2022, constructing sequences in two contiguous blocks (pre-COVID and post-COVID). The exclusion window up to December 2022—rather than an earlier cut—is justified by the observation that year-on-year rates normalise from 2023.

To ensure a fair comparison with tabular models, an XGBoost is retrained under the same exclusion protocol (XGB-noCOVID) and the original XGB trained on the full panel with dummies (XGB-Original) is also included.

5.3.3 Simple Hold-out results

Results on the twelve-month hold-out are presented in Table 5.3, ordered by MAPE from lowest to highest.

Table 5.3: Simple hold-out results (12 months), all metrics in absolute levels.

Model	RMSE	MAE	MAPE
LSTM-direct12 V2 (YoY)	89,563	68,563	4.36 %
LSTM-1step V2 (YoY)	96,269	74,103	4.71 %
XGB-noCOVID	136,752	103,011	5.83 %
XGB-Original	137,970	103,963	5.96 %
LSTM-1step V1 (levels)	197,268	140,249	8.18 %
LSTM-direct12 V1 (levels)	646,173	487,562	25.33 %

The central finding is the magnitude of the improvement introduced by the year-

on-year formulation. In the recursive variant, MAPE falls from 8.18 % to 4.71 % (42 % improvement). In the direct multi-step prediction variant, the improvement is even more spectacular: from 25.33 % to 4.36 % (83 % improvement). Same architecture, same data, same features; the only thing that changes is the representation of the target.

The asymmetry between architectures is interpretable. The direct12 variant produces all twelve values in a single forward pass and is therefore fully exposed to the level shift between training and evaluation regimes. The 1step variant iterates month by month and is partially protected because each prediction only needs to capture short-term dynamics. In the year-on-year formulation, both receive the same “methodological gift”: the structural change is absorbed by the observed denominator.

5.3.4 Expanding-window backtest

The expanding backtest over sixteen origins distributed across pre- and post-COVID blocks confirms the robustness of the finding. In the post-COVID block at a twelve-month horizon—the relevant horizon and regime for deployment—LSTM-direct12 V2 achieves a MAPE of 3.33 %, compared to 6.93 % for XGB-noCOVID, practically half. The LSTM’s precision at this long horizon is also **stable**: it oscillates between 3.33 % and 7.28 % across the four evaluated horizons (1, 3, 6 and 12 months), without catastrophic degradation, as can be seen in Figure 5.4.

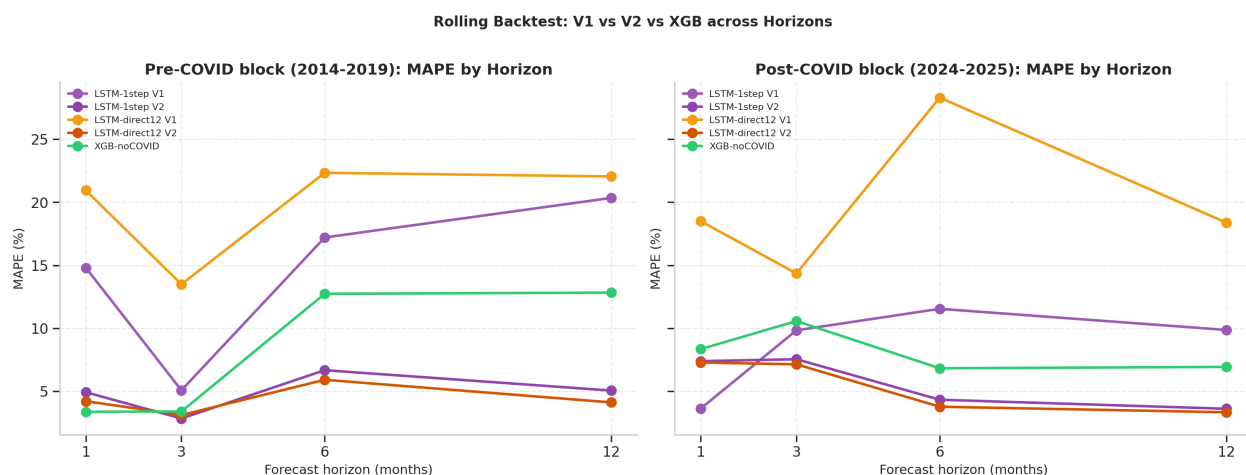


Figure 5.4: MAPE in expanding backtest by horizon, separated by pre-COVID and post-COVID blocks.

5.3.5 Diebold-Mariano test

Paired Diebold-Mariano tests at a 12-month horizon in the post-COVID block yield statistical draws for almost all pairs, due to the small number of overlapping forecasts ($n = 6$). This is a recognised limitation of the test under small samples. The decision between models is therefore not grounded in formal significance but in the **consistent direction of the differences** and their **economic magnitude**: reducing MAPE from 6.93 % to 3.33 % is a

50 % relative improvement, a difference that any analyst would consider substantial regardless of its p-value.

5.3.6 Consolidated ranking after NB07

Combining the NB05 baselines, the NB06 ML models and those of NB07, the consolidated ranking by MAPE on the twelve-month hold-out, excluding degenerate solutions, places **LSTM-direct12 V2 as the best model of the work (4.36 %)**, followed by LSTM-1step V2 (4.71 %), XGB-noCOVID (5.83 %), univariate SARIMA (6.0 %) and the tree-based models from NB06 (6–7 %).

5.3.7 Variable importance of the winning model

Unlike the tree-based models in NB06, the LSTM architecture does not expose a native variable importance measure: its internal weights are distributed non-linearly across the two recurrent layers and subsequent dense layers, and do not admit direct interpretation. To obtain a measure comparable with that reported in Figure 5.3 for the Random Forest in NB06, we apply to the winning model LSTM-direct12 V2 the **permutation importance** method (Fisher, Rudin and Dominici, 2019), a model-agnostic technique that quantifies the contribution of each variable by measuring how much the MAPE on the hold-out degrades when the temporal values of that variable are randomly shuffled within the model’s input window.

The procedure, repeated ten times per variable to estimate shuffling dispersion, operates on the twelve-month window that LSTM-direct12 V2 uses as input. A variable whose temporal permutation degrades MAPE by an appreciable amount contributes useful information to the model; a variable whose permutation leaves MAPE practically unchanged does not.

The features of LSTM-direct12 V2 comprise: five macroeconomic variables (`hicp_ratio`, `exr_pln`, `gdp_pc_origins_wavg`, `pop_origins_wavg`, `cci_DE_lag2`), one lagged air capacity variable (`seats_lag6`), twelve monthly seasonal dummies (`month_01` to `month_12`) and a linear trend variable (`trend`). Figure 5.5 reports the individual importances of each one. The acronym table in the Glossary describes each variable in detail.

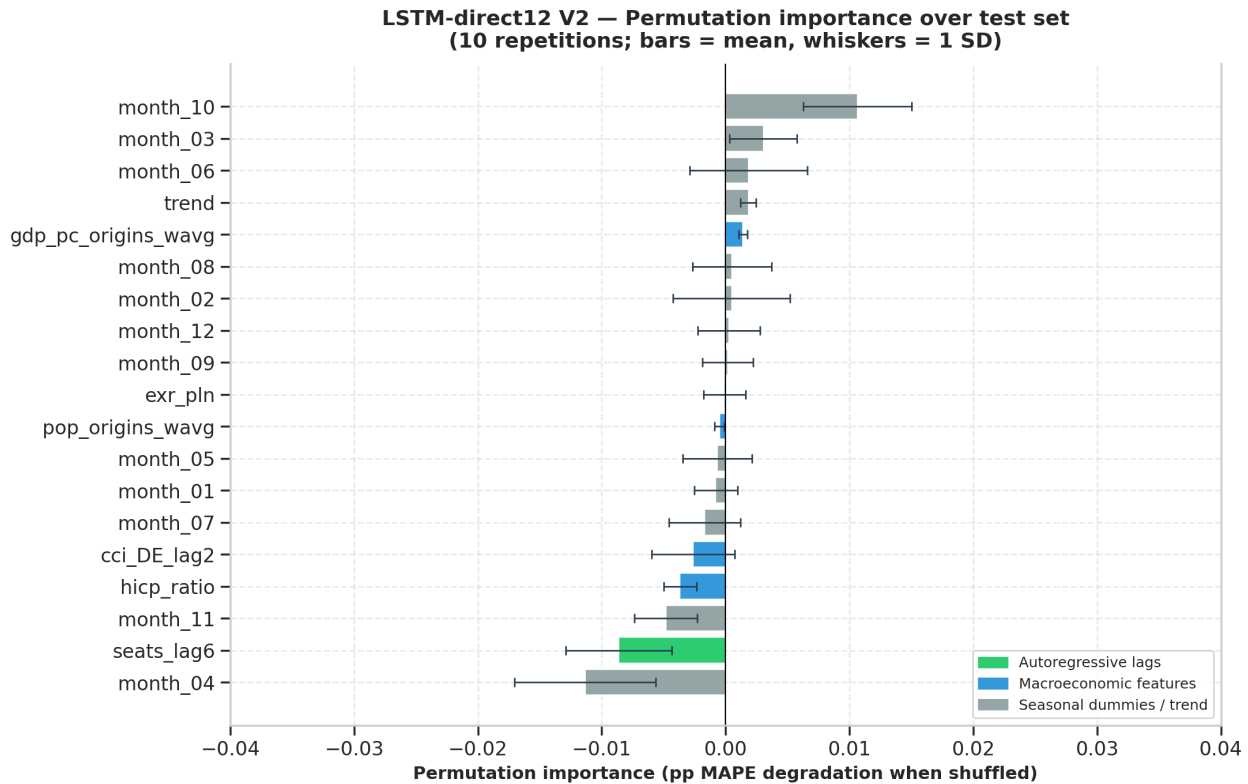


Figure 5.5: Permutation importance of LSTM-direct12 V2 on the hold-out (10 repetitions).

Reading the result. The model’s reference MAPE is 4.36 % on the hold-out. The individual degradations obtained by permuting each variable lie in the range ± 0.01 percentage points. Two distinct dimensions of this result deserve interpretation.

Regarding **statistical significance**, some variables exhibit means several times greater than their standard deviation across repetitions (`hicp_ratio`: ratio 4.0; `month_10`: 2.8; `seats_lag6`: 2.3), indicating that their effect on MAPE when permuted is not attributable solely to the random noise of the shuffling. There are therefore variables with a role distinguishable from zero in the prediction.

Regarding **economic magnitude**, however, all importances are extraordinarily small in absolute terms: the most influential variable (`month_10`) degrades MAPE by only 0.011 percentage points, which on a reference MAPE of 4.36 % represents 0.25 % in relative terms. That is, no variable contributes more than a very small fraction to the model’s aggregate performance. The presence of negative importances in variables such as `month_04` or `hicp_ratio` is likely explained by non-linear interaction between features: by randomly destroying one variable, the model may occasionally produce marginally better forecasts when that variable was contributing a noise component or redundancy with others.

The combined reading is therefore qualitative, not null: the model learns something from each variable, but each individual contribution is marginal. This is consistent with a structural property of LSTM-direct12 V2: the year-on-year transformation described in Section 5.3.1

means that the most informative variable for predicting `nights[t]` is `nights[t-12]`, which does not appear as a feature of the model because it acts as the denominator of the transformation. That is, most of the predictable variance is absorbed in the rescaling from YoY to levels, and the model only needs to learn a small seasonal and macroeconomic adjustment on top of that anchor.

This reading is fully consistent with three prior findings of the work: (i) the LSTM’s insensitivity to the IMF’s alternative macroeconomic scenarios documented in NB08, which motivated the introduction of the Peng et al. (2015) elasticity benchmark as a complementary instrument; (ii) the healthy redistribution of importance towards macroeconomic variables in the M3 specification of XGBoost in NB06 when `nights_lag1` is restricted; and (iii) the general warning by Gunter and Smeral (2016) about the dilution of macroeconomic elasticities in models with dominant autoregressive structure.

The operational consequence of this confirmation is relevant: LSTM-direct12 V2 is an excellent **point predictor**, but its use as an instrument for answering questions of the type “how much does source-market GDP influence tourism demand?” is not justified, because the model treats that variable as residual information once the seasonal anchor is controlled for. Complementarity with methods explicitly calibrated for structural sensitivity—the NB08 elasticity benchmark, or the OLS model from NB09—is not optional but **necessary** to cover the scope of analysis conditioned on macroeconomic inputs.

5.4 Prospective scenarios and causal analysis of COVID-19 (NB08)

Notebook NB08 uses the LSTM selected in NB07 to produce forecasts under alternative macroeconomic scenarios derived from the IMF’s *World Economic Outlook*, and applies the *CausalImpact* framework to estimate the causal counterfactual impact of COVID-19. The forecasting horizon covers from February 2025 to November 2026, totalling 22 months.

5.4.1 Construction of the three scenarios

The three scenarios are constructed by adjusting the annual GDP growth trajectory of source markets published by the IMF in April 2025 (IMF, 2025). The Baseline scenario uses the IMF projection without modification. The Optimistic and Pessimistic scenarios apply compound shifts of +2 percentage points and −2 percentage points respectively, from 2026 onwards. The ± 2 pp amplitude lies within the range of observable historical macroeconomic volatility—the 2008–2009 financial crisis represented approximately ± 4 points on annual growth—and is visible in the elasticity benchmark forecasts, though, as discussed below, not in the LSTM’s.

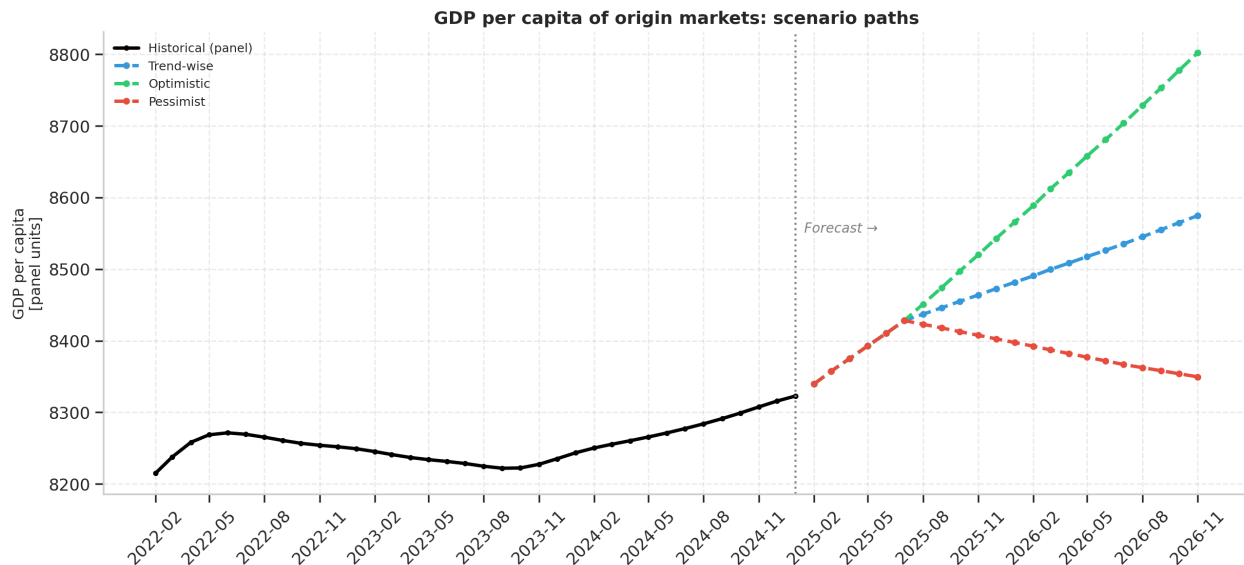


Figure 5.6: Weighted source-market GDP per capita trajectories under the three IMF scenarios.

5.4.2 Central LSTM forecast and the scenario insensitivity problem

The central forecast of LSTM-direct12 V2 places total foreign overnight stays in 2026 (January–November) at approximately 19.6 million, with a seasonal distribution concentrated in the summer months. **The three LSTM forecasts are practically indistinguishable across scenarios**, with variations below 0.1 %.

This result is not a model failure but an expected property of the NB07 design: the LSTM was calibrated for predictive accuracy, not structural sensitivity. Its forecast is dominated by the seasonal dynamics captured in `nights_lag12` and `seats_lag6`, both invariant across scenarios. Its effective income elasticity is close to zero by design. This robustness against macroeconomic noise, which in the context of NB07 was a virtue, in the context of NB08 becomes a limitation that must be documented and resolved with a complementary instrument.

5.4.3 Elasticity benchmark as sensitivity instrument

To introduce genuine sensitivity to the scenarios, the notebook applies an independent benchmark based on the average tourism income elasticity from Peng et al. (2015), equal to 2.0. Unlike the LSTM, this benchmark responds proportionally to GDP changes by mathematical construction. Applied to the IMF trajectories with ± 2 pp shifts, it produces an envelope of approximately $\pm 4\%$ around the Baseline scenario: the Optimistic adds around 800,000 annual overnight stays and the Pessimistic subtracts a similar amount. The Optimistic–Pessimistic envelope amplitude represents approximately 8 % of the annual total. Figure 5.7 synthesises the combined prospective exercise: the observed historical series, the central LSTM

forecast and the three elasticity benchmark trajectories under the Baseline, Optimistic and Pessimistic scenarios.

The two methods are complementary, not substitutes. The LSTM provides the **best central estimate**, empirically calibrated on the complete Polish panel with historical accuracy of the order of 3–5 %. The elasticity benchmark provides the **best estimate of structural sensitivity**, calibrated with an average elasticity from the world literature. The operational way of using them jointly is: the LSTM central forecast as the expected point estimate, and the benchmark envelope as a band of structural uncertainty in response to variations in the macroeconomic scenario. In absolute levels, the two methods converge within 7–15 %, which constitutes the main sanity check of the entire prospective analysis.

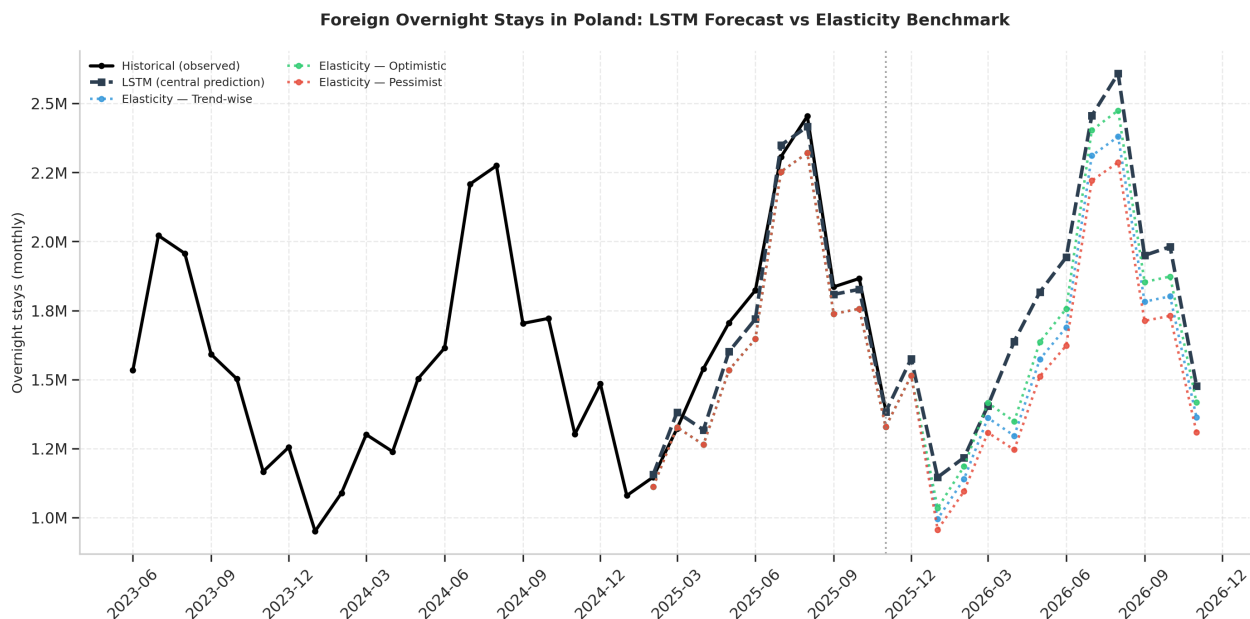


Figure 5.7: Combined LSTM and elasticity benchmark forecast under the three scenarios.

5.4.4 Causal analysis of COVID-19

The application of the *CausalImpact* framework to the foreign overnight stays series, with the intervention fixed at March 2020 and the control series formed by `seats_lag6` and seasonal dummies, yields the following results. The Bayesian model estimates that between March 2020 and November 2025 Poland lost approximately **23.6 million cumulative foreign overnight stays** relative to the no-shock counterfactual, with a 95 % credibility interval between 14.6 and 32.6 million. The average relative effect is estimated at -22.7% (95 % CI: $[-31.4\%, -14.1\%]$). The posterior probability of a real causal effect is 100 %.

This result is methodologically robust and provides a causal dimension complementary to the rest of the work, which is predictive in nature. It allows the question “*how much would Poland have grown in tourism without COVID?*” to be answered with a number and an interval, rather than with a naive comparison against the previous year that would ignore

counterfactual growth. Figure 5.8 provides a graphical visualisation of the aforementioned impact over recent years, both the collapse and the subsequent recovery.

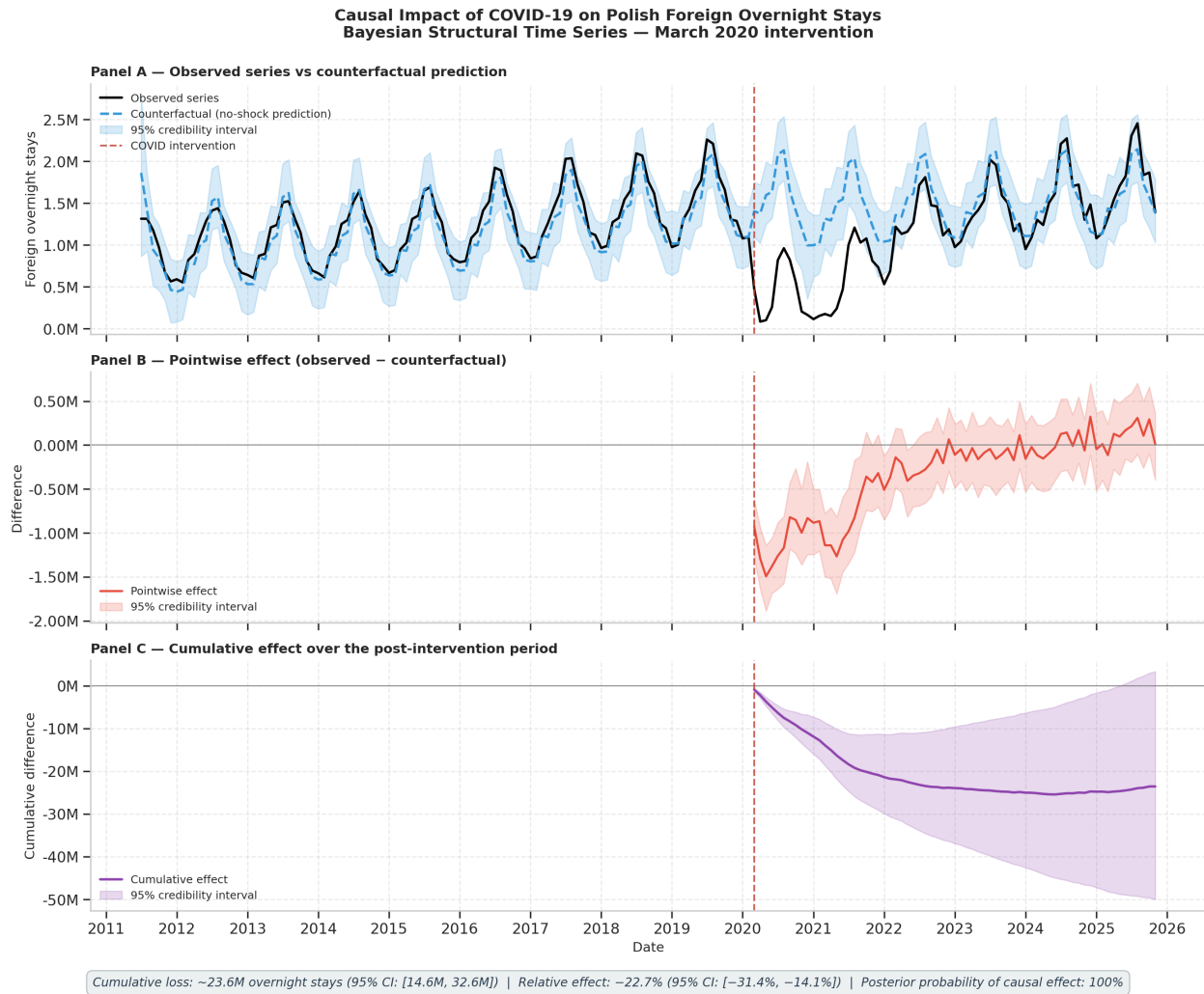


Figure 5.8: Counterfactual causal analysis of COVID-19 via CausalImpact: observed vs. counterfactual series (A), point effect (B) and cumulative effect (C).

5.5 Applied air capacity forecasting model (NB09)

Notebook NB09 develops a module oriented towards operational application in the aviation sector, in response to the dialogue with an economist with experience in the sector. The question it addresses is of an inverse nature to that of NB08: rather than predicting how many overnight stays there will be given a certain capacity and economic context, it forecasts how many seats will be offered at Polish airports given a future macroeconomic scenario.

5.5.1 Design and two parallel models

The notebook trains two parallel models with the same target variable (`seats`, total monthly seats offered at Polish airports) but with opposing philosophies.

Model A is a OLS regression in year-on-year log-differences with a single regressor: the year-on-year log-difference of the weighted source-market GDP at a 12-month lag. Its operational justification is straightforward: airlines plan capacity approximately one year in advance, so predicting seats with 12-month-lagged macro data aligns with the actual decision cycle. The estimated coefficient has a direct interpretation as an elasticity.

Model B is a non-linear XGBoost with seven lagged features, designed to maximise accuracy at the expense of interpretability. It serves as a comparator for Model A: if the flexible model does not significantly outperform the simple linear one, the linear wins on interpretability; if it surpasses it by a large margin, the flexible model is the deployment candidate.

5.5.2 Results and the honest lesson

The MAPEs on the hold-out, ordered from lowest to highest, are: Model B (XGBoost) 1.77 %, Seasonal Naïve 1.82 % and Model A (OLS) 2.01 %. The difference between the three is marginal: only 0.24 percentage points of MAPE separate the best from the worst. Three facts merit attention. First, the flexible XGBoost outperforms the Naïve by only 0.05 percentage points: the most sophisticated model in the analysis barely justifies its complexity against the trivial benchmark. Second, Model A—the interpretable OLS regression with a single macroeconomic regressor—is strictly worse than the Naïve on the hold-out, confirming that the lagged weighted GDP adds no substantial predictive information beyond the seasonal inertia captured by construction in the year-on-year formulation. Third, Model A’s formula produces a GDP coefficient of approximately 1.57 with a wide confidence interval (marginal significance, $p \approx 0.05$) and an R^2 of only 0.074, values that confirm the weak linear explanatory power of the macroeconomic variable over air capacity dynamics.

The honest reading of these results is that most of the variance in year-on-year capacity growth is explained by the inertia of the planning cycle itself (the `seats[t-12]` anchor is implicit in the year-on-year formulation), not by contemporaneous or lagged macroeconomic dynamics. This does not invalidate the model: it repositions it conceptually. The value of NB09 does not lie in predictive accuracy—where the naïve is practically equivalent—but in enabling **forecasts conditioned on macroeconomic scenarios**, which the naïve cannot do. An airline wishing to assess how a recession in its source markets would affect its capacity policy can do so with Model A; the naïve only extrapolates trend.

5.5.3 The estimated elasticity and omitted variable bias

The elasticity coefficient estimated in Model A (≈ 1.57) falls within the 1.5–2.5 range documented in the literature for the sensitivity of air capacity supply to source-market GDP. However, it must be read as a **reduced-form association**, not as a structural causal elasticity, due to the foreseeable omitted variable bias affecting the model. Exchange rates, purchasing power parities and consumer confidence are plausibly correlated with both source-market

GDP and seat supply, and their omission leaves the GDP coefficient absorbing part of their effects.

A **multivariate specification** with exchange rate, relative price and consumer confidence, all at lag 12, was empirically evaluated. Results were worse than those of Model A: the ex-COVID sample available (94 months) is too short and macroeconomic variables move within too narrow a band during the clean period for the additional regressors to provide identifying variation; they introduce parameters without adding information, which translates into worse out-of-sample accuracy. The simple specification with a single regressor is therefore, on this dataset, not only the more communicable option but also the empirically more robust one.

5.5.4 The formula as deliverable

The closed form of Model A, presentable as an operational deliverable to a sector user, is:

$$\text{seats}(t) = \text{seats}(t - 12) \cdot \exp(0.0742 + 1.5654 \cdot \Delta \log \text{GDP}_{\text{orig}}(t - 12)) \quad (5.1)$$

where $\Delta \log \text{GDP}_{\text{orig}}(t - 12)$ represents the year-on-year log-difference of the weighted source-market GDP twelve months earlier. **The forecast is not a causal explanation** and should be used as a scenario simulation instrument, not as a statement about the structural determinants of air capacity. Figure 5.9 shows the fit of the three approaches on the hold-out: the three curves are practically indistinguishable from one another, graphical evidence of the marginality of the precision differences discussed in Section 5.5.2.

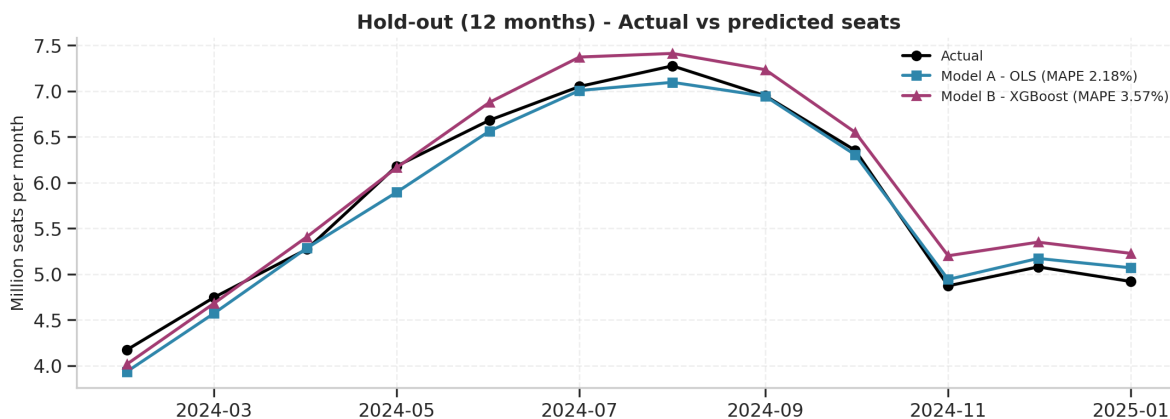


Figure 5.9: Comparison of the three air capacity forecasting approaches on the 12-month hold-out.

5.6 Final comparative synthesis

Table 5.4 summarises the main models of the work with their respective MAPEs on the twelve-month hold-out, grouped by chapter.

Table 5.4: Final consolidated ranking by MAPE.

Model	Target	Notebook	MAPE
LSTM-direct12 V2*	Overnight stays (YoY)	NB07	4.36 %
LSTM-1step V2	Overnight stays (YoY)	NB07	4.71 %
XGB-noCOVID	Overnight stays	NB07	5.83 %
XGB-Original (M3)	Overnight stays	NB07	5.96 %
SARIMA(1,1,1)(1,1,1) ₁₂	Overnight stays	NB05	6.00 %
SARIMAX(1,1,1)(1,1,1) ₁₂	Overnight stays	NB05	6.60 %
RF M3-NoLag1	Overnight stays	NB06	6.30 %
XGB M3-NoLag1	Overnight stays	NB06	7.00 %
Seasonal Naïve	Overnight stays	NB05	9.10 %
Model B (XGBoost)	Air capacity	NB09	1.77 %
Seasonal Naïve	Air capacity	NB09	1.82 %
Model A (OLS log-log)	Air capacity	NB09	2.01 %

* Best model by MAPE but with negligible structural sensitivity to macroeconomic inputs; see Section 5.3.7.

Holt-Winters is excluded from the ranking as a degenerate solution (Section 5.1). The MAPEs of the air capacity models (NB09) are notably lower than those of the overnight-stay models because the capacity series is structurally smoother and more predictable—capacity is scheduled in advance and changes less abruptly than observed tourism demand—so they are not directly comparable with the other rows.

Chapter 6

Conclusions and future work

This chapter is deliberately conceived as **self-contained**: it presents the conclusions of the work in such a way that a reader could jump directly from Chapter 2 (where the objectives are stated) to this final chapter and still understand what has been done, what has been learned and what remains to be done. Specific percentages and figures are therefore avoided here, as they belong to Chapter 5; the emphasis is on knowledge extraction and an honest assessment of achievements, limitations and possible extensions.

6.1 Conclusions

6.1.1 On the fulfilment of objectives

The work has covered the five specific objectives stated in Chapter 2 without exception. The descriptive comparative analysis of Poland relative to its source markets was completed in the initial notebooks; the key derived indicators, including the relative tourist price from the literature, were constructed through a reproducible pipeline; the modelling of the impact of each variable on tourism flows was addressed via the four-specification scheme of decreasing complexity, which empirically quantifies the contribution of each informational block once endogeneity problems are resolved; the progressive battery of predictive models was implemented covering the three families set out in the proposal—classical, machine learning and deep learning—and was enriched with Bayesian methods for causal inference; and, finally, the scenario-conditioned forecasting system was built by feeding it with IMF projections under baseline, optimistic and pessimistic trajectories, including the applied Airline Capacity Prediction module contemplated in the last specific objective of the proposal. To these five objectives was added an unplanned but substantially enriching methodological contribution: the formulation of the recurrent model’s target as a year-on-year growth rate, a decision that turned out to be one of the most relevant contributions of the project and is described in Chapter 5.

6.1.2 The main methodological contribution

If the work were to leave the community a single methodological finding, it would be the **empirical demonstration of the strength of the year-on-year formulation in recurrent networks applied to series with structural change**. The transformation, conceptually simple, decomposes the series into a stable seasonal shape—which the network learns—and a changing level—which remains in the observed denominator and is automatically reintroduced when rescaling the forecast to levels. The magnitude of the improvement obtained over the levels formulation is large enough that it cannot be attributed to chance or hyperparameter optimisation: it comes from the change in representation of the target, not from the model.

The bibliographic review carried out in the context of the work identifies several documented approaches for dealing with structural shocks in seasonal time series: formal breakpoint detection (Bai and Perron, 2003; Hall, 2025), Markov-switching regime change models (Lardic and Mignon, 2003), flexible trend decomposition with Bayesian structural methods (Doornik et al., 2021) and the treatment of the shock as a sequence of outliers in automated seasonal adjustment procedures. In the specific domain of post-COVID tourism forecasting, the consulted studies (Polyzos et al., 2021; Hsieh, 2021; Salamanis et al., 2022) respond to structural change predominantly through hybrid CNN-LSTM models or by incorporating alternative data such as Google Trends, maintaining the target formulation in absolute levels. The transformation of the recurrent model’s target to year-on-year growth rate, as implemented in the present work, has not been found documented systematically in the specific tourism forecasting literature consulted, although it shares conceptual foundations with operational practices in logistics and retail planning (Opex Analytics, 2020). The magnitude of the improvement obtained over the levels formulation is large enough that it cannot be attributed to chance or hyperparameter optimisation: it comes from the change in representation of the target, not from the model. This lesson is generalisable to other time series with structural level changes—pandemic shocks, financial crises, regulatory reforms—and constitutes a design recommendation with potential impact beyond the Polish case.

6.1.3 The distinction between prediction and explanation

A second lesson, formulated with particular clarity in notebook NB09 and summarised in a phrase that deserved highlighting during the validation phase with the academic supervisor, is that **the best prediction is not necessarily the best explanation**. A model that predicts exceptionally well does not thereby provide a causal understanding of the underlying mechanisms. The air capacity model from NB09 illustrates this principle clearly: its predictive capacity is high because the inertia of the planning cycle itself captures most of the variance; the GDP coefficient, which is the interpretable piece of the formula, exhibits marginal significance and a low R^2 . The model is valuable as a scenario simulation instrument, but should not be presented as a causal explanation of airline behaviour. This distinction, apparently subtle, is crucial for the academic honesty of the work and for its responsible use in applied contexts. As the academic supervisor emphasised during the results validation phase, “the best prediction is not necessarily the best explanation; a good prediction does not imply causality.”

6.1.4 The distinction between academic and operational objective

A third lesson, articulated throughout the work, is that the **optimal specification of a model depends on the objective pursued**. For academic inference on the determinants of tourism demand, short-term autoregressive lags are informative and should be retained, since autocorrelation is a well-documented phenomenon useful for modelling the inertia of the series. For operational prediction oriented towards decision-making, those same lags turn out to be useless because they represent information not available at the time of the decision. The four-specification scheme of decreasing complexity in notebook NB06 explicitly quantifies this cost and provides a model user with the honest figure of how much accuracy is lost in exchange for operating with available information. This principle should form part of the standard vocabulary of applied forecasting practice, and this work provides empirical evidence of its quantitative importance.

6.1.5 The complementarity between prediction and structural sensitivity

A fourth lesson, manifested when combining notebooks NB07 and NB08, is that **models calibrated for predictive accuracy tend to lose structural sensitivity** to macroeconomic variables. The work's LSTM, deliberately designed to minimise macroeconomic noise, achieves high predictive accuracy but responds minimally to scenario changes in the prospective exercise. The methodological response adopted—combining the LSTM as the central instrument with a literature elasticity benchmark as a sensitivity instrument—is defensible and produces internally consistent results (both methods converge in order of magnitude) but opens a fundamental question for future work: how to design predictive models that simultaneously maintain high accuracy and high structural sensitivity.

6.1.6 The value of complementary causal inference

The causal analysis via *CausalImpact* adds to the work a dimension that no predictive model, however sophisticated, could have provided on its own: **a quantitative estimate, with a credibility interval, of the counterfactual impact of COVID-19 on Polish tourism**. The resulting figure, expressed in millions of accumulated lost overnight stays, is not the answer to the question “*how much did tourism fall in 2020?*”—which is a naive comparison against the previous year—but to the deeper question “*how much would tourism have been in the absence of the shock, and what is the difference with what actually occurred?*”. This mode of counterfactual causal reasoning is of growing importance in applied economics and enriches the work with a component that is complementary to the dominant predictive analysis.

6.1.7 On the applied orientation of the work

The work is not only a methodological contribution; it is also a tool. The two applied modules—the prospective pipeline under IMF scenarios and the air capacity model—are designed to be used, not just read. The closed-form formula of Model A from notebook NB09, with its single interpretable parameter, is designed so that an aviation sector economist can

incorporate it into their own planning analyses without needing to understand the architectural details of the recurrent networks that complement it. This communicative versatility, which produces outputs adapted both to the academic reader and to the professional reader, is a value of the work that deserves to be highlighted.

6.2 Limitations

This section is permitted greater length than the preceding ones because honestly acknowledging limitations is one of the essential conditions of rigorous work. Limitations are grouped into five categories.

Data limitations. The main data limitation is the inability to disaggregate foreign overnight stays by country of origin, due to the fact that Eurostat dataset `tour_occ_nim` only distinguishes three aggregate categories in its residence variable. This limitation forces the modelling of the aggregate foreign overnight stays series and the incorporation of the source-market dimension only through weighted exogenous variables. If Eurostat were to publish a breakdown by country of origin at some point, it would be possible to model overnight stays by origin-destination pair, which would enable the identification of market-specific elasticities and might reveal important heterogeneities that the aggregate model cannot capture. A second data limitation is that Eurostat macroeconomic series are published with a delay of two to four months relative to the reference month, producing gaps at the end of the panel that have had to be filled via carry-forward imputation of the last published value. Although this procedure is standard for slowly changing variables, it introduces noise in the final months that cannot be strictly quantified.

Sample size limitations. After excluding the COVID period, the training set for recurrent networks is reduced to approximately 150 sequences, a modest sample for a model of LSTM complexity. The regularisers adopted—substantial dropout, early stopping on validation loss, small architecture—mitigate but do not eliminate the risk of overfitting. More significantly, the post-COVID block available at the end of the panel covers barely two years, which limits the model’s ability to learn the dynamics of the new regime. This issue will only be resolved with the passage of time, as more post-pandemic history accumulates. This limitation also affects the Diebold-Mariano tests, which in many pairs do not reach statistical significance due to the small number of overlapping forecasts.

Prospective methodology limitations. The scenarios in notebook NB08 are constructed via fixed shifts of ± 2 percentage points on the IMF baseline, a deliberate simplification that does not reflect the real probability distribution of macroeconomic risks. A more sophisticated analysis would use the IMF’s own upside and downside risk assessments published in each *World Economic Outlook* release, weighting the scenarios by their estimated probabilities. Additionally, the LSTM used in the prospective pipeline is structurally insensitive to macroeconomic scenario changes, for the reasons discussed in Chapter 5. The response adopted—complementing the LSTM with a literature elasticity benchmark—is defensible but introduces a trade-off between predictive accuracy and structural sensitivity that a single model, explicitly calibrated for both objectives, could ideally avoid.

Causal analysis limitations. The *CausalImpact* analysis of COVID uses the `seats_lag6` series and seasonal dummies as controls. The choice of the six-month lag mitigates the simultaneity problem—airlines could not adjust their monthly capacity instantaneously in response to the shock—but does not eliminate it entirely, given that capacity decisions during 2020–2022 were clearly influenced by the demand collapse. A more rigorous version of the analysis would use as counterfactual data from a comparable country not affected by the shock, which is non-trivial in the case of COVID-19 as a global pandemic. Additionally, the credibility interval of the cumulative effect reported in the work is calculated as the cumulative sum of point-wise intervals, an approximation that assumes independence between errors and tends to slightly widen the bands relative to a full Bayesian posterior simulation. The point estimate of the cumulative effect is not affected by this approximation; only the width of the interval is.

Applied model limitations. The model in notebook NB09 uses a single macroeconomic regressor (the weighted source-market GDP with a 12-month lag) and is therefore exposed to omitted variable bias, as the academic supervisor acknowledged during the validation phase. A multivariate specification including exchange rate, relative price and consumer confidence, all at lag 12, was attempted, but results were worse due to the limited variation of macroeconomic variables in the ex-COVID sample available. The GDP elasticity estimate should be interpreted as a reduced-form association, not as a structural causal elasticity. Additionally, the model has barely higher predictive power than a seasonal naïve benchmark, which limits its utility as a pure predictor and reinforces its use as a scenario simulation instrument.

6.3 Future work

Several lines of extension of the work are immediately identifiable and are listed below in approximate order of relevance.

Geographic extension of the methodological framework. The methodological scheme developed for Poland is directly applicable to other Eastern European countries with comparable dynamics, such as the Czech Republic, Hungary or Romania. A systematic cross-country comparison would allow the generalisability of the work’s conclusions to be evaluated, particularly those relating to the robustness of the year-on-year formulation and the persistence of the properties of the COVID shock.

Granularity of the target. If Eurostat were to publish a breakdown by country of origin in its overnight stays dataset, it would be possible to model origin-destination pairs with market-specific elasticities, potentially revealing important heterogeneities that the aggregate model does not capture. Alternatively, complementary sources such as aggregated mobile phone data, Google Trends searches, or reservations from platforms such as Airbnb can approximate this disaggregation if they can be consistently integrated with official statistics.

Advanced deep learning architectures. The work has been limited to the LSTM architecture due to its consolidation in the literature. Natural extensions include BiLSTM architectures, attention mechanisms, or transformer-based models applied to time series.

The warning documented in Chapter 2 about the sensitivity of recurrent networks to small sample sizes likely intensifies with more complex architectures, so their exploration should be accompanied by careful cross-validation.

Rigorous quantification of uncertainty. All models in the work produce point forecasts or, in the case of the causal analysis, Bayesian credibility intervals. A natural extension is to produce well-calibrated prediction intervals for all models, through techniques such as conformal prediction, quantile forecasting or probabilistic ensembles. This line is especially relevant in the context of the IMF prospective module, where honest quantification of uncertainty is essential for applied decision-making.

Integration of high-frequency alternative data. The models in the work use exclusively official sources with monthly frequency. The incorporation of alternative data—web searches, mobility trackers, real-time hotel reservations, meteorological data—could enrich predictive capacity, especially at short horizons where recent information is valuable. This line requires solving the nowcasting problem (predicting the present and the immediate past) with data published at different frequencies and latencies.

Air capacity models with greater granularity. The model in notebook NB09 forecasts aggregate seats at the country level. An immediate extension is the disaggregation by airport (Warsaw, Kraków, Gdańsk, Katowice) or by specific origin-destination route, which would require integrating flight scheduling data such as those published by OAG or Cirium. Airport-level granularity would be of particular utility for infrastructure planning, while route-level granularity would be essential for airline scheduling decisions.

Sustainability dimension analysis. The work has focused on the quantitative dimension of tourism (how many overnight stays, how many seats). An interesting extension would incorporate the impact dimension, measuring CO₂ emissions associated with capacity trajectories under the different scenarios. This line would connect the work with Sustainable Development Goals 12 and 13, particularly with the debate on the sustainable dimensioning of tourism growth.

6.4 Use of artificial intelligence

In accordance with the regulations of the School of Computer Engineering, the use of generative artificial intelligence tools in the preparation of this Bachelor's Thesis is explicitly declared.

Areas of use. Artificial intelligence, specifically conversational assistant language models, has been used in three main areas. First, as **programming assistance**: generation and debugging of Python code for the Jupyter notebooks, including data manipulation with `pandas`, model implementation with `scikit-learn`, `XGBoost` and `TensorFlow/Keras` for complex tasks, and modelling automation; assistance with visualisation using `matplotlib` and `seaborn`, and statistical testing with `scipy` and `statsmodels`. All decisions on what code to implement, how to design experiments and how to interpret results have been taken by the author; the AI has acted as a productivity tool, not as a decision-maker. Second, as **support in the structuring and writing** of the thesis: improvement of expository clarity,

review of coherence between chapters. All findings, methodological decisions, hypotheses, interpretations and conclusions are the exclusive responsibility of the author. Third, as **support in locating bibliographic references**: searching and verification of DOIs and editorial details. All cited references have been read, verified and incorporated by the author.

Explicitly excluded areas. Artificial intelligence has **not** taken autonomous methodological decisions. Variable selection, the definition of the four specifications in notebook NB06, the endogeneity diagnostic, the strategy for treating COVID in the LSTM, the decision to reformulate the target as a year-on-year rate, the pipeline design in NB08, the choice of the two parallel models in NB09, and all technical decisions in the work belong to the author, taken with the guidance of the academic supervisor and the external supervisor. Numerical results are the product of running the code on real data, not generated by AI.

Traceability and reproducibility. The complete code of the work is available in the project's public GitHub repository, enabling independent verification of all reported results.

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Annex A. Variable acronyms in the panel

Table 6.1 lists the acronyms of the variables used throughout the work, with a concise description of each to facilitate the reader’s consultation while reading Chapters 4 and 5.

Table 6.1: Acronyms of variables used in the panel.

Acronym	Description
<code>nights</code>	Monthly overnight stays by foreign tourists in Poland. Target variable of the work.
<code>nights_lag1</code>	Overnight stays observed one month earlier (short-term autoregressive lag).
<code>nights_lag12</code>	Overnight stays observed twelve months earlier (seasonal lag).
<code>nights_yoy</code>	Year-on-year growth of overnight stays, defined as <code>nights[t]</code> / <code>nights[t-12]</code> . Target of the LSTM in the V2 formulation.
<code>seats</code>	Total monthly available seats at Polish airports.
<code>seats_lag6</code>	Available seats six months earlier; predetermined by the standard IATA slot allocation horizon.
<code>gdp_origins_wavg</code>	Quarterly real GDP interpolated to monthly frequency, weighted by the passenger share of the nine source markets.
<code>gdp_pc_origins_wavg</code>	Weighted real GDP per capita of source markets.
<code>gdp_origins_lag3</code>	Weighted source-market GDP at a three-month lag (aligned with the cross-correlation peak).
<code>gdp_DE</code>	Real GDP of Germany, the source market with the largest share.
<code>gdp_pc_DE</code>	Real GDP per capita of Germany.
<code>pop_origins_wavg</code>	Weighted population of source markets.
<code>hicp_PL</code>	Harmonised Index of Consumer Prices for Poland (base 2015 = 100).
<code>hicp_origins_wavg</code>	Weighted average HICP of source markets.
<code>hicp_ratio</code>	Ratio between <code>hicp_PL</code> and <code>hicp_origins_wavg</code> . Indicates the relative cost of living between destination and origin.
<code>relprice_wavg</code>	Relative tourist price according to Song et al. (2010), combining HICP and exchange rates.
<code>exr_pln</code>	Bilateral euro/Polish zloty exchange rate (EUR/PLN).

(Continued on next page)

(Continued from Table 6.1)

Acronym	Description
<code>cci_DE</code>	German Consumer Confidence Indicator published by the European Commission.
<code>cci_DE_lag2</code>	CCI for Germany at a two-month lag.
<code>covid_strong</code>	Dummy variable equal to 1 between March 2020 and December 2021. Captures the acute phase of the pandemic shock.
<code>covid_post</code>	Dummy variable equal to 1 from January 2022. Captures the post-pandemic level shift.
<code>month_01,</code> <code>month_12</code>	Monthly seasonal dummies, one per calendar month. Equal to 1 if the observation belongs to the corresponding month.
<code>trend</code>	Incremental linear trend variable, equal to the number of months elapsed since the start of the panel.

Glossary

ADF (Augmented Dickey-Fuller). Statistical test for diagnosing stationarity in time series. The null hypothesis is the presence of a unit root; rejecting it implies that the series is stationary. 28, 62

BSTS (Bayesian Structural Time Series). Family of Bayesian models that decompose a time series into trend, seasonality and a regression component, with variable selection via spike-and-slab priors. 20, 62

CausalImpact Methodological framework developed by Brodersen et al. (2015) for estimating the counterfactual causal impact of an intervention on a time series, based on a BSTS model trained on the pre-intervention period. 20, 62

CCI (Consumer Confidence Indicator). Consumer confidence indicator published by the European Commission, considered a leading indicator of household spending. 62

Chain-linking (statistical chain-linking) Procedure for combining time series from different sources at a common overlap point, adjusting the levels via a scale factor and preserving the growth rates of each source outside the common period. 12, 21, 62

CRISP-DM (Cross-Industry Standard Process for Data Mining). Standard methodology for data science projects structured in six iterative phases: business understanding, data understanding, data preparation, modelling, evaluation and deployment. iv, 5, 17, 62

Diebold-Mariano test Formal statistical test for comparing the predictive accuracy of two models on a common set of forecasts. Commonly applied with the Harvey-Leybourne-Newbold correction to mitigate bias in small samples. v, 11, 12, 20, 40, 62

Dummy (indicator variable) Binary variable taking the value 1 in the presence of a condition and 0 in its absence. In this work, dummies are used to identify the COVID regime and the monthly seasonal components. 62

Endogeneity Situation in which an explanatory variable is correlated with the model's error term, typically due to reciprocal causation with the target. In this work, contemporaneous air capacity is endogenous with respect to tourism demand. iv, 5, 9, 12, 14, 28, 31, 35, 62

- Expanding-window backtest** Temporal validation scheme in which the model is retrained at successive origins across the historical series, evaluating at each origin the forecasts at multiple horizons against the observed values. v, 12, 20, 40, 62
- GDP** (Gross Domestic Product). Standard measure of the aggregate economic output of a country. 1, 62
- Granger causality** Statistical relationship whereby past values of a variable X help predict the current values of a variable Y beyond what the past values of Y already predict. Does not imply economic causality in the strict sense. 10, 62
- HICP** (Harmonised Index of Consumer Prices). Index calculated with a uniform methodology across the entire European Union in accordance with Regulation (EU) 2016/792. 62
- Hold-out** (Hold-out set). Simple validation scheme in which a final portion of the historical series is reserved as a test set on which the model trained on the remainder is evaluated. v, 12, 34, 39, 62
- KPSS** (Kwiatkowski-Phillips-Schmidt-Shin). Statistical test complementary to the ADF; rejecting the null hypothesis indicates absence of stationarity. 28, 62
- LCC** (Low-Cost Carrier). Airline whose business model is based on minimising operating costs. Ryanair and Wizz Air are the main LCCs operating in the Polish market. 7, 62
- LSTM** (Long Short-Term Memory). Recurrent neural network architecture designed to capture long-term temporal dependencies through a gate system that regulates the persistence of information through time. 10, 18, 20, 39, 62
- MAE** (Mean Absolute Error). Average of the absolute differences between forecasts and observations. 20, 62
- MAPE** (Mean Absolute Percentage Error). Accuracy metric expressed as a percentage, particularly useful for its interpretability. 20, 62
- OLS** (Ordinary Least Squares). Standard procedure for estimating linear regression models. 43, 47, 62
- ONS** (Office for National Statistics). The national statistical institute of the United Kingdom. 21, 62
- Overnight stay** Each night that a visitor (tourist) spends or pays to spend in a tourist accommodation establishment, according to the European statistical definition. It is the main target variable of the work. 62
- PIP** (Posterior Inclusion Probability). In Bayesian models with variable selection, the posterior probability that a variable is included in the true model. 62

PLN (Polish zloty). Official currency of the Republic of Poland. 28, 62

Random Forest Machine learning algorithm based on an ensemble of decision trees trained on bootstrap samples of the training set. 10, 20, 62

RMSE (Root Mean Square Error). Square root of the mean squared error. Accuracy metric that penalises large errors more heavily than the MAE. 20, 62

SARIMA / SARIMAX Seasonal autoregressive integrated moving average models, in their univariate version (SARIMA) and with exogenous variables (SARIMAX). 19, 62

SDGs Sustainable Development Goals of the United Nations, a set of seventeen global targets adopted in 2015. 15, 62

Slots (Airport slot). Authorisation to use the infrastructure of an airport at a specified time, allocated in accordance with IATA guidelines. 28, 31, 62

VIF (Variance Inflation Factor). Diagnostic metric for multicollinearity among explanatory variables. Values above 5 are considered indicative of moderate collinearity and above 10 of problematic collinearity. 30, 31, 62

WEO (World Economic Outlook). Periodic publication of the International Monetary Fund with worldwide macroeconomic projections by country. 62

XGBoost (eXtreme Gradient Boosting). Machine learning algorithm based on gradient boosting over decision trees, optimised for speed and performance. 10, 20, 62

YoY (Year-on-Year). Year-on-year change, calculated as the difference or ratio between the value in the current period and the value twelve months earlier. 26, 62